WIKIPEDIA
25 years of the free encyclopedia

Neil Armstrong

Neil Alden Armstrong (August 5, 1930 – August 25, 2012) was an American astronaut and aeronautical engineer who, as the commander of the 1969 Apollo 11 mission, became the first person to walk on the Moon. He was also a naval aviator, test pilot and university professor.

Armstrong was born and raised near Wapakoneta, Ohio. He entered Purdue University, studying aeronautical engineering, with the United States Navy paying his tuition under the Holloway Plan. He became a midshipman in 1949 and a naval aviator the following year. He saw action in the Korean War, flying the Grumman F9F Panther from the aircraft carrier USS *Essex*. After the war, he completed his bachelor's degree at Purdue and became a test pilot at the National Advisory Committee for Aeronautics (NACA) High-Speed Flight Station at Edwards Air Force Base in California. He was the project pilot on Century Series fighters and flew the North American X-15 seven times. He was also a participant in the U.S. Air Force's Man in Space Soonest and X-20 Dyna-Soar human spaceflight programs.

Armstrong joined the NASA Astronaut Corps in the second group, which was selected in 1962. He made his first spaceflight as command pilot of Gemini 8 in March 1966, becoming NASA's first civilian astronaut to fly in space. During this mission with pilot David Scott, he performed the first docking of two spacecraft; the mission was aborted after Armstrong used some of his re-entry control fuel to stabilize a dangerous roll caused by a stuck thruster. During training for Armstrong's second and last spaceflight as commander of Apollo 11, he had to eject from the Lunar Landing Research Vehicle moments before a crash.

On July 20, 1969, Armstrong and Apollo 11 Lunar Module (LM) pilot Buzz Aldrin became the first people to land on the Moon, and the next day they spent two and a half hours outside the Lunar Module

Neil Armstrong



Armstrong in 1969

Born	Neil Alden Armstrong August 5, 1930 Washington Township, Ohio, U.S.
Died	August 25, 2012 (aged 82) Fairfield, Ohio, U.S.
Education	Purdue University (BS) University of Southern California (MS)
Spouses	Janet Shearon (m. 1956; div. 1994) Carol Knight (m. 1994)
Children	3

Eagle spacecraft while Michael Collins remained in lunar orbit in the Apollo Command Module *Columbia*. When Armstrong first stepped onto the lunar surface, he famously said: "That's one small step for [a] man, one giant leap for mankind."[1][2][3][4] It was broadcast live to an estimated 530 million viewers worldwide. Apollo 11 was a major U.S. victory in the Space Race, by fulfilling a national goal proposed in 1961 by President John F. Kennedy "of landing a man on the Moon and returning him safely to the Earth" before the end of the decade. Along with Collins and Aldrin, Armstrong was awarded the Presidential Medal of Freedom by President Richard Nixon and received the 1969 Collier Trophy. President Jimmy Carter presented him with the Congressional Space Medal of Honor in 1978, he was inducted into the National Aviation Hall of Fame in 1979, and with his former crewmates received the Congressional Gold Medal in 2009.

After he resigned from NASA in 1971, Armstrong taught in the Department of Aerospace Engineering at the University of Cincinnati until 1979. He served on the Apollo 13 accident investigation and on the Rogers Commission, which investigated the Space Shuttle *Challenger* disaster. In 2012, Armstrong died due to complications resulting from coronary bypass surgery, at the age of 82.

Awards	Presidential Medal of Freedom Congressional Space Medal of Honor Congressional Gold Medal NASA Distinguished Service Medal NASA Exceptional Service Medal Air Medal (3)
	Space career
	USAF / NASA astronaut
Rank	Lieutenant, USN
Time in space	8d 14h 12m
Selection	USAF Man in Space Soonest (1958) USAF Dyna-Soar (1960) NASA Group 2 (1962)
Total EVAs	1
Total EVA time	2h 31m
Mission	Gemini 8 · Apollo 11
Mission insignia	 
Retired	August 1971

Armstrong's voice

0:00 / 0:00

Armstrong's "One small step" speech
Recorded July 20, 1969

Signature



Early life and education

Armstrong was born in rural Washington Township, in Auglaize County, Ohio,[5] on August 5, 1930, the son of Viola Louise (née Engel) and Stephen Koenig Armstrong. He was of German, English, Scots-Irish, Irish, and Scottish descent.[6][7][8] He is a descendant of Clan Armstrong.[9] He had a younger sister, June, and a younger brother, Dean. His father was an auditor for the Ohio state government,[10] and the family moved around the state repeatedly, living in 16 towns over the next 14 years.[11] Armstrong's love for flying grew during this time, having started at the age of two when his father took him to the Cleveland Air Races. When he was five or six, he experienced his first airplane flight in Warren, Ohio, when he and his father took a ride in a Ford Trimotor (also known as the "Tin Goose").[12][13] Armstrong constructed and flew model aircraft, beginning in his youth.[14]

The family's last move was in 1944 and took them back to Wapakoneta, where Armstrong attended Blume High School and took flying lessons at the Wapakoneta airfield.[15] He earned a student flight certificate on his 16th birthday, then soloed in August, all before he had a driver's license.[16] He was an active Boy Scout and earned the rank of Eagle Scout.[17] As an adult, he was recognized by the Scouts with their Distinguished Eagle Scout Award and Silver Buffalo Award.[18][19] While flying toward the Moon on July 18, 1969, he sent his regards to attendees at the National Scout jamboree in Idaho.[20] Among the few personal items that he carried with him to the Moon and back was a World Scout Badge.[21]

At age 17, in 1947, Armstrong began studying aeronautical engineering at Purdue University in West Lafayette, Indiana; he was the second person in his family to attend college. Armstrong was also accepted to the Massachusetts Institute of Technology (MIT),[22] but he resolved to go to Purdue after watching a football game between the Purdue Boilermakers and the Ohio State Buckeyes at the Ohio Stadium in 1945 in which quarterback Bob DeMoss led the Boilermakers to a sound victory over the highly regarded Buckeyes.[23] An uncle who attended MIT had also advised him that he could receive a good education without going all the way to Cambridge, Massachusetts. His college tuition was paid for under the Holloway Plan. Successful applicants committed to two years of study, followed by two years of flight training and one year of service as an aviator in the U.S. Navy, then completion of the final two years of their bachelor's degree.[22] Armstrong did not take courses in naval science, nor did he join the Naval Reserve Officers Training Corps.[24]

Naval service

Armstrong's call-up from the Navy arrived on January 26, 1949, requiring him to report to Naval Air Station Pensacola in Florida for flight training with class 5-49. After passing the medical examinations, he became a midshipman on February 24, 1949.[25] Flight training was conducted in a North American SNJ trainer, in which he soloed on September 9, 1949.[26] On March 2, 1950, he made his first aircraft carrier landing on USS *Cabot*, an achievement he considered comparable to his first solo flight.[26] He was then sent to Naval Air Station Corpus Christi in Texas for training on the Grumman F8F Bearcat, culminating in a carrier landing on USS *Wright*. On August 16, 1950, Armstrong was informed by letter that he was a fully qualified naval aviator. His mother and sister attended his graduation ceremony on August 23, 1950.[27]



Ensign Neil Armstrong on May 23, 1952

Armstrong was assigned to Fleet Aircraft Service Squadron 7 (FASRON 7) at NAS San Diego (now known as NAS North Island). On November 27, 1950, he was assigned to VF-51, an all-jet squadron, becoming its youngest officer, and made his first flight in a jet, a Grumman F9F Panther, on January 5, 1951. He was promoted to ensign on June 5, 1951, and made his first jet carrier landing on USS *Essex* two days later. On June 28, 1951, *Essex* had set sail for Korea, with VF-51 aboard to act as ground-attack aircraft. VF-51 flew ahead to Naval Air Station Barbers Point in Hawaii, where it conducted fighter-bomber training before rejoining the ship at the end of July.[28]

On August 29, 1951, Armstrong saw action in the Korean War as an escort for a photo reconnaissance plane over Songjin.[29] Five days later, on September 3, he flew armed reconnaissance over the primary transportation and storage facilities south of the village of Majon-ni, west of Wonsan. According to Armstrong, he was making a low bombing run at 350 mph (560 km/h) when 6 feet (1.8 m) of his wing was torn off after it collided with a cable that was strung across the hills as a booby trap. He was flying 500 feet (150 m) above the ground when he hit it. While there was heavy anti-aircraft fire in the area, none hit Armstrong's aircraft.[30] An initial report to the commanding officer of *Essex* said that Armstrong's F9F Panther was hit by anti-aircraft fire. The report indicated he was trying to regain control and collided with a pole, which sliced off 2 feet (0.61 m) of the Panther's right wing. Further perversions of the story by different authors added that he was only 20 feet (6.1 m) from the ground and that 3 feet (0.91 m) of his wing was sheared off.[31]



F9F-2 Panthers over Korea, with Armstrong piloting S-116 (left)

Armstrong flew the plane back to friendly territory, but because of the loss of the aileron, ejection was his only safe option. He intended to eject over water and await rescue by Navy helicopters, but his parachute was blown back over land. A jeep driven by a roommate from flight school picked him up; it is unknown what happened to the wreckage of his aircraft, F9F-2 BuNo *125122*.[32]

In all, Armstrong flew 78 missions over Korea for a total of 121 hours in the air, a third of them in January 1952, with the final mission on March 5, 1952. Of 492 U.S. Navy personnel killed in the Korean War, 27 of them were from *Essex* on this war cruise. Armstrong received the Air Medal for 20 combat missions, two gold stars for the next 40, the Korean Service Medal and Engagement Star,[33] the National Defense Service Medal, and the United Nations Korea Medal.[34]

Armstrong's regular commission was terminated on February 25, 1952, and he became an ensign in the United States Navy Reserve. On completion of his combat tour with *Essex*, he was assigned to a transport squadron, VR-32, in May 1952. He was released from active duty on August 23, 1952, but remained in the reserve, and was promoted to lieutenant (junior grade) on May 9, 1953.[34] As a reservist, he continued to fly, first with VF-724 at Naval Air Station Glenview in Illinois, and then, after moving to California, with VF-773 at Naval Air Station Los Alamitos.[35] He remained in the reserve for eight years before resigning his commission on October 21, 1960.[34]

College years

After his service with the Navy, Armstrong returned to Purdue. His grades improved after his return, lifting his final grade point average (GPA) to 4.8 out of 6.0. He pledged the Phi Delta Theta fraternity, and lived in its fraternity house. He wrote and co-directed two musicals as part of the all-student revue. The first was a version of *Snow White and the Seven Dwarfs*, co-directed with his girlfriend Joanne Alford from the Alpha Chi Omega sorority, with songs from the 1937 Walt Disney film, including "Someday My Prince Will Come"; the second was titled *The Land of Egelloc* ("college" spelled backward), with music from Gilbert and Sullivan but new lyrics.

Armstrong was chairman of the Purdue Aero Flying Club, and flew the club's aircraft, an Aeronca and a couple of Pipers, which were kept at nearby Aretz Airport in Lafayette, Indiana. Flying the Aeronca to Wapakoneta in 1954, he damaged it in a rough landing in a farmer's field, and it had to be hauled back to Lafayette on a trailer.[36] He was a baritone player in the Purdue All-American Marching Band.[37] Ten years later he was made an honorary member of Kappa Kappa Psi national band honorary fraternity.[38] Armstrong graduated with a Bachelor of Science degree in Aeronautical Engineering in January 1955.[35] In 1970, he completed his Master of Science degree in Aerospace Engineering at the University of Southern California (USC).[39] He would eventually be awarded honorary doctorates by several universities.[40]

Armstrong met Janet Elizabeth Shearon, who was majoring in home economics, at a party hosted by Alpha Chi Omega.[41] According to the couple, there was no real courtship, and neither could remember the exact circumstances of their engagement. They were married on January 28, 1956, at the Congregational Church in Wilmette, Illinois. When he moved to Edwards Air Force Base, he lived in the bachelor quarters of the base, while Janet lived in the Westwood district of Los Angeles. After one semester, they moved into a house in Antelope Valley, near Edwards AFB. Janet did not finish her degree, a fact she regretted later in life. The couple had three children.[42] In June 1961, their daughter Karen was diagnosed with diffuse intrinsic pontine glioma, a malignant tumor of the middle part of her brain stem.[43] X-ray treatment slowed its growth, but her health deteriorated to the point where she could no longer walk or talk. She died of pneumonia, related to her weakened health, on January 28, 1962, aged two.[44]

Test pilot

Following his graduation from Purdue, Armstrong became an experimental research test pilot. He applied at the National Advisory Committee for Aeronautics (NACA) High-Speed Flight Station at Edwards Air Force Base.[45] NACA had no open positions, and forwarded his application to the Lewis Flight Propulsion Laboratory in Cleveland, where Armstrong made his first test flight on March 1, 1955.[45] Armstrong's stint at Cleveland lasted only a couple of months before a position at the High-Speed Flight Station became available, and he reported for work there on July 11, 1955.[46]

On his first day, Armstrong was tasked with piloting chase planes during releases of experimental aircraft from modified bombers. He also flew the modified bombers, and on one of these missions had his first flight incident at Edwards. On March 22, 1956, he was in a Boeing B-29 Superfortress,[47] which was to air-drop a Douglas D-558-2 Skyrocket. He sat in the right-hand co-pilot seat while pilot in command, Stan Butchart, sat in the left-hand pilot seat flying the B-29.[48]

As they climbed to 30,000 feet (9 km), the number-four engine stopped and the propeller began windmilling (rotating freely) in the airstream. Despite hitting the switch that would normally stop the propeller's spinning, Butchart watched it slow before speeding up again on its own accord even faster than the others; if it spun too fast, it would break apart. Their aircraft needed to hold an airspeed of 210 mph (338 km/h) to launch its Skyrocket payload, and the B-29 could not land with the Skyrocket attached to its belly. Armstrong and Butchart brought the aircraft into a nose-down attitude to increase speed, then launched the Skyrocket. At the instant of launch, the number-four engine propeller disintegrated. Pieces of it damaged the number-three engine and hit the number-two engine. Butchart and



Armstrong, 26, as a test pilot at the NACA High-Speed Flight Station at Edwards AFB, California

Armstrong were forced to shut down the damaged number-three engine, along with the number-one engine, because of the torque it created. They made a slow, circling descent from 30,000 ft (9 km) using only the number-two engine, and landed safely.[49]

Armstrong served as project pilot on Century Series fighters, including the North American F-100 Super Sabre A and C variants, the McDonnell F-101 Voodoo, the Lockheed F-104 Starfighter, the Republic F-105 Thunderchief and the Convair F-106 Delta Dart. He also flew the Douglas DC-3, Lockheed T-33 Shooting Star, North American F-86 Sabre, McDonnell Douglas F-4 Phantom II, Douglas F5D-1 Skylancer, Boeing B-29 Superfortress, Boeing B-47 Stratojet and Boeing KC-135 Stratotanker, and was one of eight elite pilots involved in the Paresev paraglider research vehicle program.[50] Over his career, he flew more than 200 different models of aircraft.[39] His first flight in a rocket-powered aircraft was on August 15, 1957, in the Bell X-1B, to an altitude of 11.4 miles (18.3 km). On landing, the poorly designed nose landing gear failed, as had happened on about a dozen previous flights of the Bell X-1B. He flew the North American X-15 seven times,[51] including the first flight with the Q-ball system, the first flight of the number 3 X-15 airframe, and the first flight of the MH-96 adaptive flight control system.[52][53] He became an employee of the National Aeronautics and Space Administration (NASA) when it was established on October 1, 1958, absorbing NACA.[54]

Armstrong was involved in several incidents that went down in Edwards folklore or were chronicled in the memoirs of colleagues. During his sixth X-15 flight on April 20, 1962, Armstrong was testing the MH-96 control system when he flew to a height of over 207,000 feet (63 km) (the highest he flew before Gemini 8). He held up the aircraft nose during its descent to demonstrate the MH-96's g-limiting performance, and the X-15 ballooned back up to around 140,000 feet (43 km). He flew past the landing field at Mach 3 at over 100,000 feet (30 km) in altitude, and ended up 40 miles (64 km) south of Edwards. After sufficient descent, he turned back toward the landing area, and landed. It was the longest X-15 flight in both flight time and length of the ground track.[55][56]



Armstrong and X-15-1 after a research flight in 1960

Fellow astronaut Michael Collins wrote that of the X-15 pilots Armstrong "had been considered one of the weaker stick-and-rudder men, but the very best when it came to understanding the machine's design and how it operated".[57] Many of the test pilots at Edwards praised Armstrong's engineering ability. Milt Thompson said he was "the most technically capable of the early X-15 pilots". Bill Dana said Armstrong "had a mind that absorbed things like a sponge". Those who flew for the Air Force tended to have a different opinion, especially people like Chuck Yeager and Pete Knight, who did not have engineering degrees. Knight said that pilot-engineers flew in a way that was "more mechanical than it is flying", and gave this as the reason why some pilot-engineers got into trouble: Their flying skills did not come naturally.[58] Armstrong made seven flights in

the X-15 between November 30, 1960, and July 26, 1962.[59] He reached a top speed of Mach 5.74 (3,989 mph, 6,420 km/h) in the X-15-1, and left the Flight Research Center with a total of 2,400 flying hours.[60]

On April 24, 1962, Armstrong flew for the only time with Yeager. Their job, flying a T-33, was to evaluate Smith Ranch Dry Lake in Nevada for use as an emergency landing site for the X-15. In his autobiography, Yeager wrote that he knew the lake bed was unsuitable for landings after recent rains, but Armstrong insisted on flying out anyway. As they attempted a touch-and-go, the wheels became stuck and they had to wait for rescue. As Armstrong told the story, Yeager never tried to talk him out of it and they made a first successful landing on the east side of the lake. Then Yeager told him to try again, this time a bit slower. On the second landing, they became stuck, provoking Yeager to fits of laughter.[61]

On May 21, 1962, Armstrong was involved in the "Nellis Affair". He was sent in an F-104 to inspect Delamar Dry Lake in southern Nevada, again for emergency landings. He misjudged his altitude and did not realize that the landing gear had not fully extended. As he touched down, the landing gear began to retract; Armstrong applied full power to abort the landing, but the ventral fin and landing gear door struck the ground, damaging the radio and releasing hydraulic fluid. Without radio communication, Armstrong flew south to Nellis Air Force Base, past the control tower, and waggled his wings, the signal for a no-radio approach. The loss of hydraulic fluid caused the tailhook to release, and upon landing, he caught the arresting wire attached to an anchor chain, and dragged the chain along the runway.[62]

It took thirty minutes to clear the runway and rig another arresting cable. Armstrong telephoned Edwards and asked for someone to collect him. Milt Thompson was sent in an F-104B, the only two-seater available, but a plane Thompson had never flown. With great difficulty, Thompson made it to Nellis, where a strong crosswind caused a hard landing and the left main tire suffered a blowout. The runway was again closed to clear it, and Bill Dana was sent to Nellis in a T-33, but he almost landed long. The Nellis base operations office then decided that to avoid any further problems, it would be best to find the three NASA pilots ground transport back to Edwards.[62]

Astronaut career

In June 1958, Armstrong was selected for the U.S. Air Force's Man in Space Soonest program, but the Advanced Research Projects Agency (ARPA) canceled its funding on August 1, 1958, and on November 5, 1958, it was superseded by Project Mercury, a civilian project run by NASA. As a NASA civilian test pilot, Armstrong was ineligible to become one of its astronauts at this time, as selection was restricted to military test pilots.[63][64] In November 1960, he was chosen as part of the pilot consultant group for the X-20 Dyna-Soar, a military space plane under development by Boeing for the U.S. Air Force, and on March 15, 1962, he was selected by the U.S. Air Force as one of seven pilot-engineers who would fly the X-20 when it got off the design board.[65][66]



Armstrong in an early Gemini space suit

In April 1962, NASA sought applications for the second group of NASA astronauts for Project Gemini, a proposed two-man spacecraft. This time, selection was open to qualified civilian test pilots.[67] Armstrong visited the Seattle World's Fair in May 1962 and attended a conference there on space exploration that was co-sponsored by NASA. After he returned from Seattle on June 4, he applied to become an astronaut. His application arrived about a week past the June 1, 1962, deadline, but Dick Day, a flight simulator expert with whom Armstrong had worked closely at Edwards, saw the late arrival of the application and slipped it into the pile before anyone noticed.[68] At Brooks Air Force Base at the end of June, Armstrong underwent a medical exam that many of the applicants described as painful and at times seemingly pointless.[69]

NASA's Director of Flight Crew Operations, Deke Slayton, called Armstrong on September 13, 1962, and asked whether he would be interested in joining the NASA Astronaut Corps as part of what the press dubbed "the New Nine"; without hesitation, Armstrong said yes. The selections were kept secret until three days later, although newspaper reports had circulated since earlier that year that he would be selected as the "first civilian astronaut".[70] Armstrong was one of two civilian pilots selected for this group;[71] the other was Elliot See, another former naval aviator.[72] NASA selected the second group that, compared with the Mercury Seven astronauts, were younger,[69] and had more impressive academic credentials.[73] Collins wrote that Armstrong was by far the most experienced test pilot in the Astronaut Corps.[57]

Gemini program

Gemini 5

On February 8, 1965, Armstrong and Elliot See were picked as the backup crew for Gemini 5, with Armstrong as commander, supporting the prime crew of Gordon Cooper and Pete Conrad.[74] The mission's purpose was to practice space rendezvous and to develop procedures and equipment for a seven-day flight, all of which would be required for a mission to the Moon. With two other flights (Gemini 3 and Gemini 4) in preparation, six crews were competing for simulator time, so Gemini 5 was postponed. It finally lifted off on August 21.[75] Armstrong and See watched the launch at Cape Kennedy, then flew to the Manned Spacecraft Center (MSC) in Houston.[76] The mission was generally successful, despite a problem with the fuel cells that prevented a rendezvous. Cooper and Conrad practiced a "phantom rendezvous", carrying out the maneuver without a target.[77]

Gemini 8



Armstrong, 35, suiting up for Gemini 8 in March 1966

The crews for Gemini 8 were assigned on September 20, 1965. Under the normal rotation system, the backup crew for one mission became the prime crew for the third mission after, but Slayton designated David Scott as the pilot of Gemini 8.[78][79] Scott was the first member of the third group of astronauts, who was selected on October 18, 1963, to receive a prime crew assignment.[80] See was designated to command Gemini 9. Henceforth, each Gemini mission was commanded by a member of Armstrong's group, with a member of Scott's group as the pilot. Conrad would be Armstrong's backup this time, and Richard F. Gordon Jr. his

pilot.[78][79] Armstrong became the first American civilian in space. (Valentina Tereshkova of the Soviet Union had become the first civilian—and first woman—nearly three years earlier aboard Vostok 6 when it launched on June 16, 1963.[81]) Armstrong would also be the last of his group to fly in space, as See died in a T-38 crash on February 28, 1966, that also took the life of crewmate Charles Bassett. They were replaced by the backup crew of Tom Stafford and Gene Cernan, while Jim Lovell and Buzz Aldrin moved up from the backup crew of Gemini 10 to become the backup for Gemini 9,[82] and would eventually fly Gemini 12.[83]

Gemini 8 launched on March 16, 1966. It was the most complex mission yet, with a rendezvous and docking with an uncrewed Agena target vehicle, and the planned second American spacewalk (EVA) by Scott. The mission was planned to last 75 hours and 55 orbits. After the Agena lifted off at 10:00:00 EST,[84] the Titan II rocket carrying Armstrong and Scott ignited at 11:41:02 EST, putting them into an orbit from which they chased the Agena.[85] They achieved the first-ever docking between two spacecraft.[86] Contact with the crew was intermittent due to the lack of tracking stations covering their entire orbits. While out of contact with the ground, the docked spacecraft began to roll, and Armstrong attempted to correct this with the Gemini's Orbit Attitude and Maneuvering System (OAMS). Following the earlier advice of Mission Control, they undocked, but the roll increased dramatically until they were turning about once per second, indicating a problem with Gemini's attitude control. Armstrong engaged the Reentry Control System (RCS) and turned off the OAMS. Mission rules dictated that once this system was turned on, the spacecraft had to reenter at the next possible opportunity. It was later thought that damaged wiring caused one of the thrusters to stick in the on position.[87]



Recovery of Gemini 8 from the western Pacific Ocean; Armstrong sitting to the right

A few people in the Astronaut Office, including Walter Cunningham, felt that Armstrong and Scott "had botched their first mission".[88] There was speculation that Armstrong could have salvaged the mission if he had turned on only one of the two RCS rings, saving the other for mission objectives. These criticisms were unfounded; no malfunction procedures had been written, and it was possible to turn on only both RCS rings, not one or the other.[89] Gene Kranz wrote, "The crew reacted as they were trained, and they reacted wrong because we trained them wrong." The mission planners and controllers had failed to realize that when two spacecraft were docked, they must be considered one spacecraft. Kranz considered this the mission's most important lesson.[90] Armstrong was depressed that the mission was cut short,[91] canceling most mission objectives and robbing Scott of his EVA. The Agena was later reused as a docking target by Gemini 10.[92] Armstrong and Scott received the NASA Exceptional Service Medal,[93][94] and the Air Force awarded Scott the Distinguished Flying Cross as well.[95] Scott was promoted to lieutenant colonel, and Armstrong received a $678 raise in pay to $21,653 a year (equivalent to $214,864 in 2025), making him NASA's highest-paid astronaut.[91]

Gemini 11

In Armstrong's final assignment in the Gemini program, he was the back-up Command Pilot for Gemini 11. Having trained for two flights, Armstrong was quite knowledgeable about the systems and took on a teaching role for the rookie backup pilot, William Anders.[96] The launch was on September 12, 1966,[97] with Conrad and Gordon on board, who successfully completed the mission objectives, while Armstrong served as a capsule communicator (CAPCOM).[98]

Following the flight, President Lyndon B. Johnson asked Armstrong and his wife to take part in a 24-day goodwill tour of South America.[99] Also on the tour, which took in 11 countries and 14 major cities, were Dick Gordon, George Low, their wives, and other government officials. In Paraguay, Armstrong greeted dignitaries in their local language, Guarani; in Brazil he talked about the exploits of the Brazilian-born aviation pioneer Alberto Santos-Dumont.[100]

Apollo program

On January 27, 1967—the day of the Apollo 1 fire—Armstrong was in Washington, D.C., with Cooper, Gordon, Lovell and Scott Carpenter for the signing of the United Nations Outer Space Treaty. The astronauts chatted with the assembled dignitaries until 18:45, when Carpenter went to the airport, and the others returned to the Georgetown Inn, where they each found messages to phone the MSC. During these calls, they learned of the deaths of Gus Grissom, Ed White and Roger Chaffee in the fire. Armstrong and the group spent the rest of the night drinking scotch and discussing what had happened.[101]

On April 5, 1967, the same day the Apollo 1 investigation released its final report, Armstrong and 17 other astronauts gathered for a meeting with Slayton. The first thing Slayton said was, "The guys who are going to fly the first lunar missions are the guys in this room."[102] According to Cernan, only Armstrong showed no reaction to the statement. To Armstrong it came as no surprise—the room was full of veterans of Project Gemini, the only people who could fly the lunar missions. Slayton talked about the planned missions and named Armstrong to the backup crew for Apollo 9, which at that stage was planned as a medium Earth orbit test of the combined lunar module and command and service module.[103]

The crew was officially assigned on November 20, 1967.[104] For crewmates, Armstrong was assigned Lovell and Aldrin, from Gemini 12. After design and manufacturing delays of the lunar module (LM), Apollo 8 and 9 swapped prime and backup crews. Based on the normal crew rotation, Armstrong would command Apollo 11,[103] with one change: Collins on the Apollo 8 crew began experiencing trouble with his legs. Doctors diagnosed the problem as a bony growth between his fifth and sixth vertebrae, requiring surgery.[105] Lovell took his place on the Apollo 8 crew, and, when Collins recovered, he joined Armstrong's crew.[106]

To give the astronauts practice piloting the LM on its descent, NASA commissioned Bell Aircraft to build two Lunar Landing Research Vehicles (LLRV), later augmented with three Lunar Landing Training Vehicles (LLTV). Nicknamed the "Flying Bedsteads", they simulated the Moon's one-sixth gravity using a turbofan engine to support five-sixths of the craft's weight. On May 6, 1968, 100 feet (30 m) above the ground, Armstrong's controls started to degrade and the LLRV began rolling.[107] He ejected safely before the vehicle struck the ground and burst into flames. Later analysis suggested that if he had ejected half a second later, his parachute would not have opened in time. His only injury was from biting his tongue. The LLRV was completely destroyed.[108] Even though he was nearly killed, Armstrong maintained that without the LLRV and LLTV, the lunar landings would not have been successful, as they gave commanders essential experience in piloting the lunar landing craft.[109]



Armstrong descends to the ground on a parachute after ejecting from Lunar Landing Research Vehicle 1.

In addition to the LLRV training, NASA began lunar landing simulator training after Apollo 10 was completed. Aldrin and Armstrong trained for a variety of scenarios that could develop during a real lunar landing.[110] They also received briefings from geologists at NASA.[111]

Apollo 11



The Apollo 11 crew: Armstrong, Michael Collins, and Buzz Aldrin.

After Armstrong served as backup commander for Apollo 8, Slayton offered him the post of commander of Apollo 11 on December 23, 1968, as Apollo 8 orbited the Moon.[112] According to Armstrong's 2005 biography, Slayton told him that although the planned crew was Commander Armstrong, Lunar Module Pilot Buzz Aldrin, and Command Module Pilot Michael Collins, he was offering Armstrong the chance to replace Aldrin with Jim Lovell. After thinking it over for a day, Armstrong told Slayton he would stick with Aldrin, as he had no difficulty working with him and thought Lovell deserved his own command. Replacing Aldrin with Lovell would have made Lovell the lunar module pilot, unofficially the lowest ranked member, and Armstrong could not justify placing Lovell, the commander of Gemini 12, in the number 3 position of the crew.[113] The crew of Apollo 11 was assigned on January 9, 1969, as Armstrong, Collins, and Aldrin, with Lovell, Anders, and Fred Haise as the backup crew.[114]

According to Chris Kraft, a March 1969 meeting among Slayton, George Low, Bob Gilruth, and Kraft determined that Armstrong would be the first person on the Moon, in part because NASA management saw him as a person who did not have a large ego. A press conference on April 14, 1969, gave the design of the LM cabin as the reason for Armstrong's being first; the hatch opened inwards and to the right, making it difficult for the LM pilot, on the right-hand side, to exit first. At the time of their meeting, the four men did not know about the hatch consideration. The first knowledge of the meeting outside the small group came when Kraft wrote his book.[115][116] Methods of circumventing this difficulty existed, but it is not known if these were considered at the time. Slayton added, "Secondly, just on a pure protocol basis, I figured the commander ought to be the first guy out ... I changed it as soon as I found they had the time line that showed that. Bob Gilruth approved my decision."[117]

Voyage to the Moon

A Saturn V rocket launched Apollo 11 from Launch Complex 39A at the Kennedy Space Center on July 16, 1969, at 13:32:00 UTC (09:32:00 EDT local time).[118] Armstrong's wife Janet and two sons watched from a yacht moored on the Banana River.[119] During the launch, Armstrong's heart rate peaked at 110 beats per minute.[120] He found the first stage the loudest, much noisier than the Gemini 8 Titan II launch. The Apollo command module was relatively roomy compared with the Gemini spacecraft. None

of the Apollo 11 crew suffered space sickness, as some members of previous crews had. Armstrong was especially glad about this, as he had been prone to motion sickness as a child and could experience nausea after long periods of aerobatics.[121]



Armstrong in the lunar module after the completion of the EVA

Apollo 11's objective was to land safely on the Moon, rather than to touch down at a precise location. Three minutes into the lunar descent, Armstrong noted that craters were passing about two seconds too early, which meant the Lunar Module *Eagle* would probably touch down several miles (kilometres) beyond the planned landing zone.[122] As the *Eagle*'s landing radar acquired the surface, several computer error alarms sounded. The first was a code 1202 alarm, and even with their extensive training, neither Armstrong nor Aldrin knew what this code meant. They promptly received word from CAPCOM Charles Duke in Houston that the alarms were not a concern; the 1202 and 1201 alarms were caused by executive overflows in the lunar module guidance computer. In 2007, Aldrin said the overflows were caused by his own counter-checklist choice of leaving the docking radar on during the landing process, causing the computer to process unnecessary radar data. When it did not have enough time to execute all tasks, the computer dropped the lower-priority ones, triggering the alarms. Aldrin said he decided to leave the radar on in case an abort was necessary when re-docking with the Apollo command module; he did not realize it would cause the processing overflows.[123]

When Armstrong noticed they were heading toward a landing area that seemed unsafe, he took manual control of the LM and attempted to find a safer area. This took longer than expected, and longer than most simulations had taken.[124] For this reason, Mission Control was concerned that the LM was running low on fuel.[125] On landing, Aldrin and Armstrong believed they had 40 seconds of fuel left, including the 20 seconds' worth which had to be saved in the event of an abort.[126] During training, Armstrong had, on several occasions, landed with fewer than 15 seconds of fuel; he was also confident the LM could survive a fall of up to 50 feet (15 m). Post-mission analysis showed that at touchdown there were 45 to 50 seconds of propellant burn time left.[127]



Armstrong lands the Lunar Module *Eagle* on the Moon, July 20, 1969.

The landing on the surface of the Moon occurred several seconds after 20:17:40 UTC on July 20, 1969.[128] One of three 67-inch (170 cm) probes attached to three of the LM's four legs made contact with the surface, a panel light in the LM illuminated, and Aldrin called out, "Contact light." Armstrong shut the engine off and said, "Shutdown." As the LM settled onto the surface, Aldrin said, "Okay, engine stop"; then they both called out some post-landing checklist items. After a 10-second pause, Duke acknowledged the landing with, "We copy you down, *Eagle*." Armstrong confirmed the landing to Mission Control and the world with the words, "Houston, Tranquility Base here. The *Eagle* has landed." Aldrin and Armstrong celebrated with a brisk handshake and pat on the back. They then returned to the

checklist of contingency tasks, should an emergency liftoff become necessary.[129][130][131] After Armstrong confirmed touch down, Duke re-acknowledged, adding a comment about the flight crew's relief: "Roger, Tranquility. We copy you on the ground. You got a bunch of guys about to turn blue. We're breathing again. Thanks a lot."[126] During the landing, Armstrong's heart rate ranged from 100 to 150 beats per minute.[132]

First Moon walk

The flight plan called for a crew rest period before leaving the module, but Armstrong asked for this to be moved to earlier in the evening, Houston time. When he and Aldrin were ready to go outside, *Eagle* was depressurized, the hatch was opened, and Armstrong made his way down the ladder.[133] At the bottom of the ladder, while standing on a Lunar Module landing pad, Armstrong said, "I'm going to step off the LM now". He turned and set his left boot on the lunar surface at 02:56 UTC July 21, 1969,[134] then said, "That's one small step for [a] man, one giant leap for mankind."[135] The exact time of Armstrong's first step on the Moon is unclear.[136]



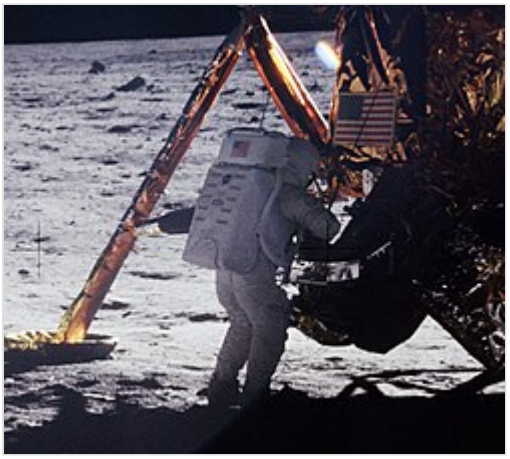
Armstrong describes the lunar surface.

Armstrong prepared his famous epigram on his own.[137] In a post-flight press conference, he said that he chose the words "just prior to leaving the LM."[138] In a 1983 interview in *Esquire* magazine, he explained to George Plimpton: "I always knew there was a good chance of being able to return to Earth, but I thought the chances of a successful touch down on the moon surface were about even money—fifty–fifty ... Most people don't realize how difficult the mission was. So it didn't seem to me there was much point in thinking of something to say if we'd have to abort landing."[137] In 2012, his brother Dean Armstrong said that Neil showed him a draft of the line months before the launch.[139] Historian Andrew Chaikin, who interviewed Armstrong in 1988 for his book *A Man on the Moon*, disputed that Armstrong claimed to have conceived the line during the mission.[140]

Recordings of Armstrong's transmission do not provide evidence for the indefinite article "a" before "man", though NASA and Armstrong insisted for years that static obscured it. Armstrong stated he would never make such a mistake, but after repeated listenings to recordings, he eventually conceded he must have dropped the "a".[135] He later said he "would hope that history would grant me leeway for dropping the syllable and understand that it was certainly intended, even if it was not said—although it might actually have been".[141] There have since been claims and counter-claims about whether acoustic analysis of the recording reveals the presence of the missing "a";[135][142] Peter Shann Ford, an Australian computer programmer, conducted a digital audio analysis and claims that Armstrong did say "a man", but the "a" was inaudible due to the limitations of communications technology of the time.[135][143][144] Ford and James R. Hansen, Armstrong's authorized biographer, presented these findings to Armstrong and NASA representatives, who conducted their own analysis.[145] Armstrong found Ford's analysis "persuasive".[146][147] Linguists David Beaver and Mark Liberman wrote of their skepticism of Ford's claims on the blog Language Log.[148] A 2016 peer-reviewed study again concluded Armstrong had included the article.[149] NASA's transcript continues to show the "a" in parentheses.[150]

When Armstrong made his proclamation, Voice of America was rebroadcast live by the BBC and many other stations worldwide. An estimated 530 million people viewed the event,[151] 20 percent out of a world population of approximately 3.6 billion.[152][153]

> Q: Did you misspeak?
>
> A: There isn't any way of knowing.
>
> Q: Several sources say you did.
>
> A: I mean, there isn't any way of *my* knowing. When I listen to the tape, I can't hear the 'a', but that doesn't mean it wasn't there, because that was the fastest VOX ever built. There was no mike-switch—it was a voice-operated key or VOX. In a helmet you find you lose a lot of syllables. Sometimes a short syllable like 'a' might not be transmitted. However, when I listen to it, I can't hear it. But the 'a' is implied, so I'm happy if they just put it in parentheses.
>
> —*Omni*, June 1982, p. 126



Armstrong on the Moon

About 19 minutes after Armstrong's first step, Aldrin joined him on the surface, becoming the second human to walk on the Moon. They began their tasks of investigating how easily a person could operate on the lunar surface. Armstrong unveiled a plaque commemorating the flight, and with Aldrin, planted the flag of the United States. Although Armstrong had wanted the flag to be draped on the flagpole,[154] it was decided to use a metal rod to hold it horizontally.[155] However, the rod did not fully extend, leaving the flag with a slightly wavy appearance, as if there were a breeze.[156] Shortly after the flag planting, President Richard Nixon spoke to them by telephone from his office. He spoke for about a minute, after which Armstrong responded for about thirty seconds.[157] In the Apollo 11 photographic record, there are only five images of Armstrong partly shown or reflected. The mission was planned to the minute, with the majority of photographic tasks performed by Armstrong with the single Hasselblad camera.[158]

After helping to set up the Early Apollo Scientific Experiment Package, Armstrong went for a walk to what is now known as East Crater, 65 yards (59 m) east of the LM, the greatest distance traveled from the LM on the mission. His final task was to remind Aldrin to leave a small package of memorial items to Soviet cosmonauts Yuri Gagarin and Vladimir Komarov, and Apollo 1 astronauts Grissom, White and Chaffee.[159] The Apollo 11 EVA lasted two and a half hours.[160] Each of the subsequent five landings was allotted a progressively longer EVA period; the crew of Apollo 17 spent over 22 hours exploring the lunar surface.[160] In a 2010 interview, Armstrong explained that NASA limited their Moon walk because they were unsure how the space suits would cope with the Moon's extremely high temperature.[161]

Return to Earth

After they re-entered the LM, the hatch was closed and sealed. While preparing for liftoff, Armstrong and Aldrin discovered that, in their bulky space suits, they had broken the ignition switch for the ascent engine; using part of a pen, they pushed in the circuit breaker to start the launch sequence.[162] The *Eagle* then continued to its rendezvous in lunar orbit, where it docked with *Columbia*, the command and service module. The three astronauts returned to Earth and splashed down in the Pacific Ocean, to be picked up by the USS *Hornet*.[163]



The Apollo 11 crew and President Richard Nixon during the post-mission quarantine period

After being released from an 18-day quarantine to ensure that they had not picked up any infections or diseases from the Moon, the crew was feted across the United States and around the world as part of a 38-day "Giant Leap" tour.[164]

The tour began on August 13, when the three astronauts spoke and rode in ticker-tape parades in their honor in New York and Chicago, with an estimated six million attendees.[165][166] On the same evening an official state dinner was held in Los Angeles to celebrate the flight, attended by members of Congress, 44 governors, the Chief Justice of the United States, and ambassadors from 83 nations. President Nixon and Vice President Agnew presented each astronaut with a Presidential Medal of Freedom.[165][167]


New York City ticker tape parade, August 13, 1969

After the tour Armstrong took part in Bob Hope's 1969 USO show, primarily to Vietnam.[168] In May 1970, Armstrong traveled to the Soviet Union to present a talk at the 13th annual conference of the International Committee on Space Research; after arriving in Leningrad from Poland, he traveled to Moscow where he met Premier Alexei Kosygin. Armstrong was the first westerner to see the supersonic Tupolev Tu-144 and was given a tour of the Yuri Gagarin Cosmonaut Training Center, which he described as "a bit Victorian in nature".[169] At the end of the day, he was surprised to view a delayed video of the launch of Soyuz 9 as it had not occurred to Armstrong that the mission was taking place, even though Valentina Tereshkova had been his host and her husband, Andriyan Nikolayev, was on board.[170]

Life after Apollo

Teaching

Shortly after Apollo 11, Armstrong stated that he did not plan to fly in space again.[171] He was appointed Deputy Associate Administrator for Aeronautics for the Office of Advanced Research and Technology at ARPA, served in the position for a year, then resigned from it and NASA in 1971.[172] He accepted a

teaching position in the Department of Aerospace Engineering at the University of Cincinnati,[173] having chosen Cincinnati over other universities, including his alma mater Purdue, because Cincinnati had a small aerospace department,[174] and said he hoped the faculty there would not be annoyed that he came straight into a professorship with only a USC master's degree.[175] He began his master's degree while stationed at Edwards years before, and completed it after Apollo 11 by presenting a report on various aspects of Apollo, instead of a thesis on the simulation of hypersonic flight.[176]



Valentina Tereshkova, the first woman in space, presenting a badge to Neil Armstrong, Star City, Soviet Union, June 1970

At Cincinnati, Armstrong was University Professor of Aerospace Engineering. He took a heavy teaching load, taught core classes, and created two graduate-level classes: aircraft design and experimental flight mechanics. He was considered a good teacher, and a tough grader. His research activities during this time did not involve his work at NASA, as he did not want to give the appearance of favoritism; he later regretted the decision. After teaching for eight years, Armstrong resigned in 1980. When the university changed from an independent municipal university to a state school, bureaucracy increased. He did not want to be a part of the faculty collective bargaining group, so he decided to teach half-time. According to Armstrong, he had the same amount of work but received half his salary. In 1979, less than 10% of his income came from his university salary. Employees at the university did not know why he left.[176]

NASA commissions

In 1970, after an explosion aboard Apollo 13 aborted its lunar landing, Armstrong was part of Edgar Cortright's investigation of the mission. He produced a detailed chronology of the flight. He determined that a 28-volt thermostat switch in an oxygen tank, which was supposed to have been replaced with a 65-volt version, led to the explosion. Cortright's report recommended the entire tank be redesigned at a cost of $40 million. Many NASA managers, including Armstrong, opposed the recommendation, since only the thermostat switch had caused the problem. They lost the argument, and the tanks were redesigned.[177]

In 1986, President Ronald Reagan asked Armstrong to join the Rogers Commission investigating the Space Shuttle *Challenger* disaster. Armstrong was made vice chairman of the commission and held private interviews with contacts he had developed over the years to help determine the cause of the disaster. He helped limit the committee's recommendations to nine, believing that if there were too many, NASA would not act on them.[178]

Armstrong was appointed to a fourteen-member commission by President Reagan to develop a plan for American civilian spaceflight in the 21st century. The commission was chaired by former NASA administrator Dr. Thomas O. Paine, with whom Armstrong had worked during the Apollo program. The group published a book titled *Pioneering the Space Frontier: The Report on the National Commission on Space*, recommending a permanent lunar base by 2006, and sending people to Mars by 2015. The recommendations were largely ignored, overshadowed by the *Challenger* disaster.[179]

Armstrong and his wife attended the memorial service for the victims of the Space Shuttle *Columbia* disaster in 2003, at the invitation of President George W. Bush.[180]



Michael Collins, President George W. Bush, Neil Armstrong, and Buzz Aldrin during celebrations of the 35th anniversary of the Apollo 11 flight, July 21, 2004

Business activities

After Armstrong retired from NASA in 1971, he acted as a spokesman for several businesses. The first company to successfully approach him was Chrysler, for whom he appeared in advertising starting in January 1979. Armstrong thought they had a strong engineering division, and they were in financial difficulty. He later acted as a spokesman for other American companies, including General Time Corporation and the Bankers Association of America.[181] He acted as a spokesman for only American companies.[182]

In addition to his duties as a spokesman, he also served on the board of directors of several companies. The first company board Armstrong joined was Gates Learjet, chairing their technical committee. He flew their new and experimental jets and even set a climb and altitude record for business jets. Armstrong became a member of Cincinnati Gas & Electric Company's board in 1973. They were interested in nuclear power and wanted to increase the company's technical competence. He served on the board of Taft Broadcasting, also based in Cincinnati. Armstrong joined the board of solid rocket booster Thiokol in 1989, after previously serving on the Rogers Commission which found that the Space Shuttle *Challenger* was destroyed due to a defect in the Thiokol-manufactured solid rocket boosters. When Armstrong left the University of Cincinnati, he became the chairman of Cardwell International Ltd., a company that manufactured drilling rigs. He served on additional aerospace boards, first United Airlines in 1978, and later Eaton Corporation in 1980. He was asked to chair the board of directors for a subsidiary of Eaton, AIL Systems. He chaired the board through the company's 2000 merger with EDO Corporation, until his retirement in 2002.[183][184]

North Pole and Ecuador expeditions

In 1976, Stanley Hall of Scotland arranged one of the most expensive large-scale cave explorations in modern history to investigate Cueva de los Tayos in Ecuador. For the expedition, over a hundred people were recruited, including members of the Scottish Black Watch and the Royal Highland Fusiliers regiment, Ecuadorian military personnel, and various experts in different fields, including Armstrong, then a professor of aerospace engineering at the University of Cincinnati, as Honorary President of the expedition.[185][186]

In 1985, professional expedition leader Mike Dunn organized a trip to take men he deemed the "greatest explorers" to the North Pole. The group included Armstrong, Edmund Hillary, Hillary's son Peter, Steve Fossett, and Patrick Morrow. They arrived at the Pole on April 6, 1985. He did not inform the media of the trip, preferring to keep it private.[187]

Public profile



Armstrong in 1999

Armstrong's family described him as a "reluctant American hero".[188][189][190] He kept a low profile later in his life, leading to the belief that he was a recluse.[191][192] Recalling Armstrong's humility, John Glenn, the first American to orbit Earth, told CNN: "[Armstrong] didn't feel that he should be out huckstering himself. He was a humble person, and that's the way he remained after his lunar flight, as well as before."[193] Armstrong turned down most requests for interviews and public appearances. Michael Collins said in his book *Carrying the Fire* that when Armstrong moved to a dairy farm to become a college professor, it was like he "retreated to his castle and pulled up the drawbridge". Armstrong found this amusing, and said, "... those of us that live out in the hinterlands think that people that live inside the Beltway are the ones that have the problems."[194]

Andrew Chaikin says in *A Man on the Moon* that Armstrong kept a low profile but was not a recluse, citing his participation in interviews, advertisements for Chrysler, and hosting a cable television series.[195] Between 1991 and 1993, he hosted *First Flights with Neil Armstrong*, an aviation history documentary series on A&E.[194] In 2010, Armstrong voiced the character of Dr. Jack Morrow in *Quantum Quest: A Cassini Space Odyssey*,[196] an animated educational sci-fi adventure film initiated by JPL/NASA through a grant from Jet Propulsion Lab.[197]

Armstrong guarded the use of his name, image, and famous quote. When it was launched in 1981, MTV wanted to use his quote in its station identification, with the American flag replaced with the MTV logo, but he refused the use of his voice and likeness.[198] He sued Hallmark Cards in 1994, when they used his name, and a recording of the "one small step" quote, in a Christmas ornament without his permission. The lawsuit was settled out of court for an undisclosed sum, which Armstrong donated to Purdue.[199][200]

For many years, he wrote letters congratulating new Eagle Scouts on their accomplishment, but decided to quit the practice in the 1990s because he felt the letters should be written by people who knew the scout. (In 2003, he received 950 congratulation requests.) This contributed to the myth of his reclusiveness.[201] Armstrong used to autograph everything except first day covers. Around 1993, he found out his signatures were being sold online, and that most of them were forgeries, and stopped giving autographs.[192]

Other activities

Some former astronauts, including Glenn and Apollo 17's Harrison Schmitt, sought political careers after leaving NASA. Armstrong was approached by groups from both the Democratic and Republican parties but declined the offers. He supported states' rights and opposed the U.S. acting as the "world's policeman".[202]

When Armstrong applied at a local Methodist church to lead a Boy Scout troop in the late 1950s, he gave his religious affiliation as "deist".[203] His mother later said that his religious views caused her grief and distress in later life, as she was a Christian.[204] Upon his return from the Moon, Armstrong gave a speech in front of the U.S. Congress in which he thanked them for giving him the opportunity to see some of the



Armstrong speaking in February 2012, six months before his death, on the 50th anniversary of John Glenn's first spaceflight

"grandest views of the Creator".[205][206] In the early 1980s, he was the subject of a hoax claiming that he converted to Islam after hearing the call to prayer while walking on the Moon. Indonesian singer Suhaemi wrote a song called "Gema Suara Adzan di Bulan" ("The Resonant Sound of the Call to Prayer on the Moon") which described Armstrong's supposed conversion, and the song was widely discussed by Jakarta news outlets in 1983.[207] Similar hoax stories were seen in Egypt and Malaysia. In March 1983, the U.S. State Department responded by issuing a message to embassies and consulates in Muslim countries saying that Armstrong had not converted to Islam.[208] The hoax surfaced occasionally for the next three decades. Part of the confusion arose from the similarity between the names of the country of Lebanon, which has a majority Muslim population, and Armstrong's longtime residence in Lebanon, Ohio.[208]

In 1972, Armstrong visited the Scottish town of Langholm, the traditional seat of Clan Armstrong. He was made the first freeman of the burgh, and happily declared the town his home.[209] To entertain the crowd, the Justice of the Peace read from an unrepealed archaic 400-year-old law that required him to hang any Armstrong found in the town.[210]

Armstrong flew light aircraft for pleasure. He enjoyed gliders and before the Moon flight had earned a gold badge with two diamonds from the International Gliding Commission. He continued to fly engineless aircraft well into his 70s.[211]

While working on his farm in November 1978, Armstrong jumped off the back of his grain truck and caught his wedding ring in its wheel, tearing the tip off his left ring finger. He collected the severed tip, packed it in ice, and had surgeons reattach it at a nearby hospital in Louisville, Kentucky.[212] In February 1991, he suffered a mild heart attack while skiing with friends at Aspen, Colorado.[213]

Armstrong and his first wife, Janet, separated in 1990 and divorced in 1994 after 38 years of marriage.[214][215] He met his second wife, Carol Held Knight, at a golf tournament in 1992, when they were seated together at breakfast. She said little to Armstrong, but he called her two weeks later to ask what she was doing. She replied that she was cutting down a cherry tree, and he arrived at her house 35 minutes later to help. They were married in Ohio on June 12, 1994, and had a second ceremony at San Ysidro Ranch in California. They lived in Indian Hill, Ohio.[216][217] Through his marriage to Carol, he was the father-in-law of future New York Mets general manager Brodie Van Wagenen.

In May 2005, Armstrong became involved in a legal dispute with Mark Sizemore, his barber of 20 years. After cutting Armstrong's hair, Sizemore sold some of it to a collector for $3,000 without Armstrong's knowledge or permission.[218] Armstrong threatened legal action against Sizemore unless he returned the hair or donated the proceeds to a charity of Armstrong's choosing. Sizemore, unable to retrieve the hair, donated the proceeds to charity.[219][220]

Illness and death

On August 7, 2012, Armstrong underwent bypass surgery at Mercy Health Fairfield Hospital in Fairfield, Ohio, to relieve coronary artery disease.[221][222] Although he was reportedly recovering well,[223] he developed complications and died on August 25.[224][225] President Barack Obama issued a statement memorializing Armstrong as "among the greatest of American heroes—not just of his time, but of all time",[226][227] and added that Armstrong had carried the aspirations of the United States' citizens and had delivered "a moment of human achievement that will never be forgotten".[228]



Photograph of Armstrong as a boy at his family memorial service in Indian Hill, Ohio, near Cincinnati, on August 31, 2012

Armstrong's family released a statement describing him as a "reluctant American hero [who had] served his nation proudly, as a navy fighter pilot, test pilot, and astronaut ... While we mourn the loss of a very good man, we also celebrate his remarkable life and hope that it serves as an example to young people around the world to work hard to make their dreams come true, to be willing to explore and push the limits, and to selflessly serve a cause greater than themselves. For those who may ask what they can do to honor Neil, we have a simple request. Honor his example of service, accomplishment and modesty, and the next time you walk outside on a clear night and see the moon smiling down at you, think of Neil Armstrong and give him a wink."[229]



Armstrong's burial at sea on September 14, 2012

Buzz Aldrin called Armstrong "a true American hero and the best pilot I ever knew", and said he was disappointed that they would not be able to celebrate the 50th anniversary of the Moon landing together in 2019.[230][231] Michael Collins said, "He was the best, and I will miss him terribly."[232][233] NASA administrator Charles Bolden said, "As long as there are history books, Neil Armstrong will be included in them, remembered for taking humankind's first small step on a world beyond our own".[234][235]

A tribute was held for Armstrong on September 13, at Washington National Cathedral, whose Space Window depicts the Apollo 11 mission and holds a sliver of Moon rock amid its stained-glass panels.[236] In attendance were Armstrong's Apollo 11 crewmates, Collins and Aldrin; Gene Cernan, the Apollo 17 mission commander and last man to walk on the Moon; and former senator and astronaut John Glenn, the first American to orbit the Earth. In his eulogy, Charles Bolden praised Armstrong's "courage, grace, and humility". Cernan recalled Armstrong's low-fuel approach to the Moon: "When the gauge says empty, we all know there's a gallon or two left in the tank!" Diana Krall sang the song "Fly Me to the Moon".

External videos

Memorial service for Armstrong, Washington National Cathedral, September 13, 2012 (https://www.c-span.org/video/?308141-1/memorial-service-neil-armstrong), C-SPAN

Collins led prayers. David Scott spoke, possibly for the first time, about an incident during their Gemini 8 mission: minutes before the hatch was to be sealed, a small chip of dried glue fell into the latch of his harness and prevented it from being buckled, threatening to abort the mission. Armstrong then called on Conrad to solve the problem, which he did, and the mission proceeded. "That happened because Neil Armstrong was a team player—he always worked on behalf of the team."[236] Congressman Bill Johnson from Armstrong's home state of Ohio led calls for President Barack Obama to authorize a state funeral in Washington D.C. Throughout his lifetime, Armstrong shunned publicity and rarely gave interviews. Mindful that Armstrong would have objected to a state funeral, his family opted to have a private funeral in Cincinnati.[237] On September 14, Armstrong's cremated remains were scattered in the Atlantic Ocean from the USS *Philippine Sea*.[238] Flags were flown at half-staff on the day of Armstrong's funeral.[239]

In July 2019, after observations of the 50th anniversary of the Moon landing, *The New York Times* reported on details of a medical malpractice suit Armstrong's family had filed against Mercy Health–Fairfield Hospital, where he died. When Armstrong appeared to be recovering from his bypass surgery, nurses removed the wires connected to his temporary pacemaker. He began to bleed internally and his blood pressure dropped. Doctors took him to the hospital's catheterization laboratory, and only later began operating. Two of the three physicians who reviewed the medical files during the lawsuit called this a serious error, saying surgery should have begun immediately; experts the *Times* talked to, while qualifying their judgment by noting that they were unable to review the specific records in the case, said that taking a patient directly to the operating room under those circumstances generally gave them the highest chance of survival.[221]

The family ultimately settled for $6 million in 2014. Letters included with the 93 pages of documents sent to the *Times* by an unknown person[240] show that his sons intimated to the hospital, through their lawyers, that they might discuss what happened to their father publicly at the 45th anniversary observances in 2014. The hospital, fearing the bad publicity that would result from being accused of negligently causing the death of a revered figure such as Armstrong, agreed to pay as long as the family never spoke about the suit or the settlement.[221] Armstrong's wife, Carol, was not a party to the lawsuit. She reportedly felt that her husband would have been opposed to taking legal action.[241]

Legacy

When Pete Conrad of Apollo 12 became the third man to walk on the Moon, on November 19, 1969, his first words referenced Armstrong. The shorter of the two, when Conrad stepped from the LM onto the surface he proclaimed "Whoopie! Man, that may have been a small one for Neil, but that's a long one for me."[242]

Armstrong received many honors and awards, including the Presidential Medal of Freedom (with distinction) from President Nixon,[165][243] the Cullum Geographical Medal from the American Geographical Society,[244] and the Collier Trophy from the National Aeronautic Association (1969);[245] the NASA Distinguished Service Medal[246] and the Dr. Robert H. Goddard Memorial Trophy (1970);[247] the Sylvanus Thayer Award by the United States Military Academy (1971);[248] the Congressional Space Medal of Honor from President Jimmy Carter (1978);[93] the Wright Brothers Memorial Trophy from the National Aeronautic Association (2001);[249] and a Congressional Gold Medal (2011).[250]

Armstrong was elected as member into the National Academy of Engineering in 1978 for contributions to aerospace engineering, scientific knowledge, and exploration of the universe as an experimental test pilot and astronaut.[251] He was elected to the American Philosophical Society in 2001.[252]



Armstrong gives an acceptance speech after being inducted into the Naval Aviation Hall of Honor at the National Naval Aviation Museum in Pensacola, Florida.

Armstrong and his Apollo 11 crewmates were the 1999 recipients of the Langley Gold Medal from the Smithsonian Institution.[253] On April 18, 2006, he received NASA's Ambassador of Exploration Award.[254] The Space Foundation named Armstrong as a recipient of its 2013 General James E. Hill Lifetime Space Achievement Award.[255] Armstrong was also inducted into the Aerospace Walk of Honor,[256][257] the International Space Hall of Fame,[258] National Aviation Hall of Fame, and the United States Astronaut Hall of Fame.[259][260] He was awarded his Naval Astronaut badge in a ceremony on board the aircraft carrier USS *Dwight D. Eisenhower* on March 10, 2010, in a ceremony attended by Lovell and Cernan.[261]


President Barack Obama poses with the Apollo 11 crew on the 40th anniversary of the Apollo 11 lunar landing, July 20, 2009: Buzz Aldrin, Michael Collins, and Neil Armstrong.

The lunar crater Armstrong, 31 miles (50 km) from the Apollo 11 landing site, and asteroid 6469 Armstrong are named in his honor.[262] There are more than a dozen elementary, middle and high schools named for Armstrong in the United States,[263] and many places around the world have streets, buildings, schools, and other places named for him or Apollo.[264] The Armstrong Air and Space Museum, in Armstrong's hometown of Wapakoneta,[265] and the Neil Armstrong Airport in New Knoxville, Ohio, are named after him.[266] The mineral armstrongite is named after him,[267] and the mineral armalcolite is named, in part, after him.[268]

In October 2004 Purdue University named its new engineering building Neil Armstrong Hall of Engineering;[269] the building was dedicated on October 27, 2007, during a ceremony at which Armstrong was joined by fourteen other Purdue astronauts.[270] The NASA Dryden Flight Research Center was renamed the NASA Neil A. Armstrong Flight Research Center in 2014.[271] NASA has separately renamed the Plum Brook Station test site at Glenn Research Center as Neil A. Armstrong Test Facility. In September 2012, the U.S. Navy named the first *Armstrong*-class vessel RV *Neil Armstrong*. Delivered to the Navy on September 23, 2015, it is a modern oceanographic research platform supporting a wide range of activities by academic groups.[272] In 2019, the College of Engineering at Purdue University celebrated the 50th anniversary of Neil Armstrong's walk on the Moon by launching the Neil Armstrong Distinguished Visiting Fellows Program, which brings highly accomplished scholars and practitioners to the college to catalyze collaborations with faculty and students.[273]

External videos

Armstrong's authorized biography, *First Man: The Life of Neil A. Armstrong*, was published in 2005. For many years, he turned down biography offers from authors such as Stephen Ambrose and James A. Michener but agreed to work with James R. Hansen after reading one of Hansen's other biographies.[274] He recalled his initial concerns about the Apollo 11 mission, when he had believed there was only a 50% chance of landing on the Moon. "I was elated, ecstatic and extremely surprised that we were successful".[275] A film adaptation of the book, starring Ryan Gosling and directed by Damien Chazelle, was released in October 2018.[276]

Presentation by James Hansen on *First Man*, November 9, 2005 (https://www.c-span.org/video/?189961-1/first-man-life-neil-a-armstrong), C-SPAN

In July 2018, Armstrong's sons consigned his collection of memorabilia to Heritage Auctions, launching a series of sales that including his Boy Scout cap, and various flags and medals flown on his space missions. A series of auctions held November 1–3, 2018, realized $5,276,320 (~$6.65 million in 2025). As of July 2019, the auction sales totaled $16.7 million.[241] Two fragments of wood from the propeller and four pieces of fabric from the wing of the 1903 *Wright Flyer* that Armstrong took to the Moon fetched between $112,500 and $275,000 each,[277][278] as well as his Apollo 11 flown gold medal, which fetched $2.06 million in a separate Heritage sale.[279] Armstrong's wife, Carol, has not put any of his memorabilia up for sale.[241]



Ohio's state quarter depicts Armstrong and the Wright brothers' Wright Flyer III.

Armstrong donated his papers to Purdue. Along with posthumous donations by his widow Carol, the collection consists of over 450 boxes of material. In May 2019, she donated two 25-by-24-inch (640 by 610 mm) pieces of fabric from the *Wright Flyer*, along with his correspondence related to them.[280][281]

In a 2010 *Space Foundation* survey, Armstrong was ranked as the number-one most popular space hero;[282] and in 2013, *Flying* magazine ranked him number one on its list of 51 Heroes of Aviation.[283] The press often asked Armstrong for his views on the future of spaceflight. In 2005, he said that a human mission to Mars would be easier than the lunar challenge of the 1960s. In 2010, he made a rare public criticism of the decision to cancel the Ares I launch vehicle and the Constellation Moon landing program.[284] In an open letter also signed by fellow Apollo veterans Lovell and Cernan, he said, "For The United States, the leading space faring nation for nearly half a century, to be without carriage to low Earth orbit and with no human exploration capability to go beyond Earth orbit for an indeterminate time into the future, destines our nation to become one of second or even third rate stature".[285] On November 18, 2010, aged 80, he said in a speech during the *Science & Technology Summit* in the Hague, Netherlands, that he would offer his services as commander on a mission to Mars if he were asked.[286]

The planetarium at Altoona Area High School in Altoona, Pennsylvania is named after Armstrong and is home to a Space Race museum.[287] A campsite in Camp Sandy Beach at Yawgoog Scout Reservation in Rockville, Rhode Island, is named in his honor, a nod to his Scouting career.

Armstrong was named the class exemplar for the Class of 2019 at the U.S. Air Force Academy.[288]

Notes

1. "Armstrong's famous 'one small step' quote—explained" (https://whyy.org/articles/armstrongs-famous-one-small-step-quote-explained/). *WHYY-FM*. July 14, 2019. Retrieved July 13, 2023.
2. "July 20, 1969: One Giant Leap For Mankind" (https://www.nasa.gov/mission_pages/apollo/apollo11.html). *NASA*. July 20, 2019. Retrieved August 13, 2023.
3. Armstrong, Neil (July 16, 1999). "[Press conference with Neil Armstrong]" (https://history.nasa.gov/ap11ann/pressconf.htm). *NASA History Division*. Retrieved August 13, 2023.
4. Stamm, Amy (July 17, 2019). " "One Small Step for Man" or "a Man"?" (https://airandspace.si.edu/stories/editorial/one-small-step-man-or-man). *National Air and Space Museum*. Retrieved August 13, 2023.
5. Butler County Department of Health (August 6, 1930). "English: This is a faithful digital scan of the official birth certificate of Neil Alden Armstrong, issued by the Butler County Department of Health on March 31, 2025. Certified by local county registrar, Kathy Ripley" (https://commons.wikimedia.org/wiki/File:Neil_Alden_Armstrong_-_Original_birth_certificate.jpg). Retrieved April 15, 2025.
6. "Distant Irish relatives mourn moonwalker Neil Armstrong" (https://www.irishcentral.com/news/distant-irish-relatives-mourn-moonwalker-neil-armstrong-167530445-237525201). August 27, 2012.
7. Hansen 2005, pp. 13, 20.
8. Coleman, Maureen (August 28, 2012). "A Giant Leap For An Ulsterman" (https://www.belfasttelegraph.co.uk/news/northern-ireland/a-giant-leap-for-an-ulsterman-tributes-to-neil-armstrong-the-astronaut-who-was-first-to-walk-on-the-moon-28785887.html). *The Belfast Telegraph*. Retrieved November 14, 2018.
9. Harvey, Ian (April 15, 2019). "Neil Armstrong's Last Name Posed a Problem in his Ancestral Scottish Hometown" (https://www.thevintagenews.com/2019/04/15/armstrong/). *thevintagenews*. Retrieved September 4, 2022.
10. "Neil Armstrong grants rare interview to accountants organization" (http://www.cbc.ca/news/technology/neil-armstrong-grants-rare-interview-to-accountants-organization-1.1289392). CBC News. May 24, 2012. Archived (https://web.archive.org/web/20180107163955/http://www.cbc.ca/news/technology/neil-armstrong-grants-rare-interview-to-accountants-organization-1.1289392) from the original on January 7, 2018. Retrieved April 8, 2018.
11. Hansen 2005, p. 29.
12. "Project Apollo: Astronaut Biographies" (https://history.nasa.gov/ap11ann/astrobios.htm#Armstrong). NASA. Archived (https://web.archive.org/web/20110428105817/http://history.nasa.gov/ap11ann/astrobios.htm#Armstrong) from the original on April 28, 2011. Retrieved May 12, 2011.
13. Hansen 2005, p. 45. "According to a volunteer group in Warren, Ohio that had worked through the 2000s to turn the Warren Airport into a historical exhibit, the date of Neil's inaugural flight was July 26, 1936. If that date is correct, Neil was still only five when he experienced his first airplane ride, his sixth birthday not coming for ten more days."
14. Haughn, Rachelle (June 2019). "One Small Step With Model Aviation" (https://www.modelaviation.com/one-small-step). *Model Aviation*. Retrieved July 29, 2025.
15. Hansen 2005, pp. 49–50.
16. Koestler-Grack 2010, p. 14.
17. Hansen 2012, p. 38.
18. Airgood, Glenn (February 16, 1973). "1st Man on the Moon Gets National Eagle Award" (https://www.newspapers.com/clip/23385802/the_morning_call/). *The Morning Call*. Allentown, Pennsylvania. p. 5 – via Newspapers.com.
19. "Silver Buffalo Award Winners 1979–1970" (https://www.scouting.org/awards/silver-buffalo/previous-winners/1979-1970/). Boy Scouts of America. Retrieved September 2, 2018.

20. "Apollo 11 – Day 3, part 2: Entering Eagle – Transcript" (https://history.nasa.gov/afj/ap11fj/09day3-entering-eagle.html). NASA. April 11, 2010. Retrieved February 14, 2022. "I'd like to say hello to all my fellow Scouts and Scouters at Farragut State Park in Idaho having a National Jamboree there this week; and Apollo 11 would like to send them best wishes". Capsule communicator Charles Duke replied: "Thank you, Apollo 11. I'm sure that, if they didn't hear that, they'll get the word through the news. Certainly appreciate that."
21. "World Scouting salutes Neil Armstrong" (https://web.archive.org/web/20150904004807/http://oldsite.scout.org/en/information_events/news/world_scouting_salutes_neil_armstrong). World Organization of the Scout Movement. Archived from the original (http://oldsite.scout.org/en/information_events/news/world_scouting_salutes_neil_armstrong) on September 4, 2015. Retrieved July 27, 2015.
22. Hansen 2005, pp. 55–56.
23. "The untold story of how Neil Armstrong chose Purdue" (https://web.archive.org/web/20190704035141/https://www.wlfi.com/content/news/The-untold-story-of-how-Neil-Armstrong-chose-Purdue-512192321.html). *wlfi.com*. Archived from the original (https://www.wlfi.com/content/news/The-untold-story-of-how-Neil-Armstrong-chose-Purdue-512192321.html) on July 4, 2019. Retrieved July 5, 2019.
24. Hansen 2005, p. 58.
25. Hansen 2005, pp. 68–69.
26. Hansen 2005, p. 71.
27. Hansen 2005, pp. 76–79.
28. Hansen 2005, pp. 79–85.
29. Hansen 2005, p. 90.
30. Hansen 2005, p. 94.
31. Hansen 2005, pp. 92–93.
32. Hansen 2005, pp. 95–96.
33. Hansen 2005, p. 112.
34. "Ex-Lieutenant (junior grade) Neil Alden Armstrong, U.S. Naval Reserve, Transcript of Naval Service" (https://www.history.navy.mil/content/dam/nhhc/research/histories/bios/Armstrong-Neil/Armstrong_redacted_Redacted.pdf) (PDF). United States Navy. March 27, 1967. Archived (https://web.archive.org/web/20170506025424/https://www.history.navy.mil/content/dam/nhhc/research/histories/bios/Armstrong-Neil/Armstrong_redacted_Redacted.pdf) (PDF) from the original on May 6, 2017. Retrieved February 28, 2018.
35. Hansen 2005, p. 118.
36. Hansen 2005, pp. 61–62.
37. "Purdue mourns alumnus Neil Armstrong" (https://www.purdue.edu/newsroom/releases/2012/Q3/purdue-mourns-alumnus-neil-armstrong.html). Purdue University. August 25, 2012. Archived (https://web.archive.org/web/20171216125803/http://www.purdue.edu/newsroom/releases/2012/Q3/purdue-mourns-alumnus-neil-armstrong.html) from the original on December 16, 2017. Retrieved April 2, 2018.
38. "Purdue Bands launch $2 million fund-raising campaign" (https://www.purdue.edu/uns/html3month/1997/970425.Bands.campaign.html). Purdue University. April 25, 1997. Retrieved July 10, 2018.
39. "Biographical Data: Neil A. Armstrong" (https://web.archive.org/web/20171204041951/https://www.jsc.nasa.gov/Bios/htmlbios/armstrong-na.html). NASA. August 2012. Archived from the original (https://www.jsc.nasa.gov/Bios/htmlbios/armstrong-na.html) on December 4, 2017. Retrieved April 7, 2018.
40. "Biography: Neil A. Armstrong" (https://www.nasa.gov/centers/glenn/about/bios/neilabio.html). NASA (Glenn Research Center). March 2008. Archived (https://web.archive.org/web/20110526004733/http://www.nasa.gov/centers/glenn/about/bios/neilabio.html) from the original on May 26, 2011. Retrieved May 16, 2011.

41. Hansen 2005, p. 62.
42. Hansen 2005, pp. 124–128.
43. Chillag, Amy; Higgins, Cole (March 1, 2019). "Girl, 7, Fighting Rare Cancer Gets Pics of Dogs from Well-Wishers" (https://edition.cnn.com/2019/03/01/health/iyw-girl-cancer-dog-photos-trnd/index.html). CNN. Retrieved November 24, 2019.
44. Hansen 2005, pp. 161–164.
45. Hansen 2005, pp. 119–120.
46. Hansen 2005, p. 130.
47. Hansen 2005, p. 134.
48. Creech, Gray (July 15, 2004). "From the Mojave to the Moon: Neil Armstrong's Early NASA Years" (https://web.archive.org/web/20110630200654/http://www.nasa.gov/missions/research/neil_armstrong.html). NASA. Archived from the original (https://www.nasa.gov/missions/research/neil_armstrong.html) on June 30, 2011. Retrieved May 17, 2011.
49. Hansen 2005, pp. 134–136.
50. Hansen 2005, pp. 136–138.
51. Hansen 2005, p. 145.
52. Evans, Michelle (2013). "The X-15 Rocket Plane: Flight Log" (http://www.mach25media.com/Resources/X15FlightLog.pdf) (PDF). Mach 25 Media. pp. 22, 25. Archived (https://web.archive.org/web/20180413002147/http://www.mach25media.com/Resources/X15FlightLog.pdf) (PDF) from the original on April 13, 2018. Retrieved February 28, 2018.
53. Hansen 2005, p. 147.
54. "T. Keith Glennan" (https://web.archive.org/web/20170214234112/http://www.hq.nasa.gov/office/pao/History/Biographies/glennan.html). NASA. Archived from the original (https://www.hq.nasa.gov/office/pao/History/Biographies/glennan.html) on February 14, 2017. Retrieved March 4, 2018.
55. Hansen 2005, pp. 178–184.
56. Klesius, Mike (May 20, 2009). "Neil Armstrong's X-15 flight over Pasadena" (https://www.smithsonianmag.com/air-space-magazine/neil-armstrongs-x-15-flight-over-pasadena-59458462/). *Smithsonian Air & Space Magazine*. Retrieved January 25, 2023.
57. Collins 2001, pp. 314.
58. Hansen 2005, pp. 138–139.
59. Jenkins 2000, pp. 118–121.
60. Hansen 2005, p. 210.
61. Hansen 2005, pp. 184–189.
62. Hansen 2005, pp. 189–192.
63. Burgess 2013, pp. 17–18.
64. Reichhardt, Tony (August–September 2000). "First Up?" (http://www.airspacemag.com/space-exploration/first_up.html). *Air & Space*. Retrieved February 28, 2018.
65. Hansen 2005, pp. 171–173.
66. Burgess 2013, pp. 19–21.
67. Burgess 2013, pp. 4–6.
68. Hansen 2005, pp. 193–195.
69. Burgess 2013, pp. 29–30.
70. Hansen 2005, pp. 201–202.
71. Burgess 2013, pp. 64–66.
72. "Elliot M. See, Jr" (https://web.archive.org/web/20110513235725/http://www.jsc.nasa.gov/Bios/htmlbios/see-em.html). NASA. Archived from the original (https://www.jsc.nasa.gov/Bios/htmlbios/see-em.html) on May 13, 2011. Retrieved May 19, 2011.

73. Burgess 2013, p. 54.
74. Reichl 2016, p. 78.
75. Hacker & Grimwood 2010, pp. 255–256.
76. Hansen 2005, p. 239.
77. Hacker & Grimwood 2010, pp. 257–258.
78. Hansen 2005, p. 240.
79. Hacker & Grimwood 2010, pp. 523–529.
80. "14 New Astronauts Introduced at Press Conference" (https://web.archive.org/web/20161222121909/http://www.jsc.nasa.gov/history/roundups/issues/63-10-30.pdf) (PDF). *Space News*. Vol. 3, no. 1. October 30, 1963. Archived from the original (https://www.jsc.nasa.gov/history/roundups/issues/63-10-30.pdf) (PDF) on December 22, 2016. Retrieved February 28, 2018.
81. "Valentina Vladimirovna Tereshkova (Born March 6, 1937)" (https://web.archive.org/web/20150904004808/http://www.yarregion.ru/eng/Pages/famous_people_Valentina_Vladimirovna_Tereshkova.aspx). Yaroslavl Regional Government. Archived from the original (http://www.yarregion.ru/eng/Pages/famous_people_Valentina_Vladimirovna_Tereshkova.aspx) on September 4, 2015. Retrieved July 27, 2015.
82. Hacker & Grimwood 2010, pp. 323–325.
83. Cunningham 2010, p. 258.
84. Hansen 2005, pp. 242–244.
85. Hacker & Grimwood 2010, p. 526.
86. "March 16, 1966: Gemini's First Docking of Two Spacecraft in Earth Orbit" (https://www.nasa.gov/image-feature/march-16-1966-geminis-first-docking-of-two-spacecraft-in-earth-orbit). NASA. March 16, 2016. Retrieved April 30, 2018.
87. Merritt, Larry (March 2006). "The abbreviated flight of Gemini 8" (https://web.archive.org/web/20110812222659/http://bts.boeing.com/news/frontiers/archive/2006/march/i_history.html). Boeing. Archived from the original (http://bts.boeing.com/news/frontiers/archive/2006/march/i_history.html) on August 12, 2011. Retrieved May 14, 2011.
88. Cunningham 2010, pp. 111–112.
89. Hansen 2005, pp. 270–271.
90. Kranz 2000, p. 174.
91. Hansen 2005, p. 274.
92. Hacker & Grimwood 2010, pp. 321–322.
93. *Agency Awards Historical Recipient List* (https://searchpub.nssc.nasa.gov/servlet/sm.web.Fetch/Agency_Awards_Historical_Recipient_List.pdf?rhid=1000&did=2120817&type=released) (PDF), NASA, archived (https://web.archive.org/web/20161202083252/https://searchpub.nssc.nasa.gov/servlet/sm.web.Fetch/Agency_Awards_Historical_Recipient_List.pdf?rhid=1000&did=2120817&type=released) (PDF) from the original on December 2, 2016, retrieved February 28, 2018
94. "Serious Problem in Space" (https://www.newspapers.com/clip/31060661/the_times_recorder/). *The Times Recorder*. Zanesville, Ohio. United Press International. March 27, 1966. p. 8 – via Newspapers.com.
95. "Valor awards for David Randolph Scott" (https://valor.militarytimes.com/recipient.php?recipientid=143831). Military Times Hall of Valor. Archived (https://web.archive.org/web/20180301225001/https://valor.militarytimes.com/recipient.php?recipientid=143831) from the original on March 1, 2018. Retrieved February 28, 2018.
96. Hansen 2005, pp. 292–293.

97. "Gemini-XI" (https://web.archive.org/web/20120201213341/http://science.ksc.nasa.gov/history/gemini/gemini-xi/gemini-xi.html). NASA (Kennedy Space Center). August 25, 2000. Archived from the original (https://science.ksc.nasa.gov/history/gemini/gemini-xi/gemini-xi.html) on February 1, 2012. Retrieved July 24, 2010.
98. Hansen 2005, pp. 294–296.
99. Hansen 2005, pp. 296–297.
100. Hansen 2005, pp. 298–299.
101. Lovell & Kluger 2000, pp. 24–25.
102. Cernan & Davis 1999, p. 165.
103. Hansen 2005, pp. 312–313.
104. Brooks et al. 2009, p. 374.
105. Collins 2001, pp. 288–289.
106. Cunningham 2010, p. 109.
107. Hansen 2005, p. 330.
108. Kraft 2001, p. 312.
109. Hansen 2005, p. 334.
110. Chaikin 1994, p. 171.
111. Chaikin 1994, p. 179.
112. Nelson 2009, p. 17.
113. Hansen 2005, p. 338.
114. Collins 2001, pp. 312–313.
115. Kraft 2001, pp. 323–324.
116. Hansen 2005, pp. 365–373.
117. Cortright 1975, p. 160.
118. Orloff 2000, p. 92.
119. Hansen 2005, p. 2.
120. Hansen 2005, p. 410.
121. Hansen 2005, pp. 411–412.
122. Smith 2005, p. 11.
123. Hansen 2005, pp. 459–465.
124. Chaikin 1994, p. 199.
125. Chaikin 1994, p. 198.
126. Chaikin 1994, p. 200.
127. Manned Spacecraft Center 1969, pp. 9-23–9-24.
128. Jones, Eric M. "The First Lunar Landing, time 109:45:40" (https://web.archive.org/web/20171225232200/https://history.nasa.gov/alsj/a11/a11.landing.html). *Apollo 11 Surface Journal*. NASA. Archived from the original (https://www.history.nasa.gov/alsj/a11/a11.landing.html) on December 25, 2017. Retrieved March 4, 2018. That was the time of probe contact; the exact time of landing is difficult to determine, because Armstrong said the landing was "very gentle" and "It was hard to tell when we were on."
129. Jones, Eric M. (September 15, 2017). "The First Lunar Landing, time 1:02:45" (https://web.archive.org/web/20171225232200/https://history.nasa.gov/alsj/a11/a11.landing.html). *Apollo 11 Surface Journal*. NASA. Archived from the original (https://www.history.nasa.gov/alsj/a11/a11.landing.html) on December 25, 2017. Retrieved November 30, 2007.

130. Jones, Eric M. "Mission Transcripts, Apollo 11 AS11 PA0.pdf" (https://web.archive.org/web/20080917232517/http://www.jsc.nasa.gov/history/mission_trans/AS11_PAO.PDF) (PDF). *Apollo 11 Surface Journal*. NASA. Archived from the original (https://www.jsc.nasa.gov/history/mission_trans/AS11_PAO.PDF) (PDF) on September 17, 2008. Retrieved November 30, 2007.
131. Jones, Eric M. "Apollo 11 Mission Commentary 7-20-69 CDT 15:15 – GET 102:43 – TAPE 307/1" (https://web.archive.org/web/20171108072819/https://history.nasa.gov/alsj/a11/a11transcript_pao.htm). *Apollo 11 Surface Journal*. NASA. Archived from the original (https://history.nasa.gov/alsj/a11/a11transcript_pao.htm) on November 8, 2017.
132. Manned Spacecraft Center 1969, p. 12-1.
133. Cortright 1975, p. 215.
134. Harland 1999, p. 23.
135. Mikkelson, Barbara; Mikkelson, David (October 2006). "One Small Misstep: Neil Armstrong's First Words on the Moon" (http://www.snopes.com/quotes/onesmall.asp). *Snopes.com*. Retrieved September 19, 2009.
136. Stern, Jacob (July 23, 2019). "One Small Controversy About Neil Armstrong's Giant Leap" (https://www.theatlantic.com/science/archive/2019/07/tiny-imprecision-heart-apollo-11/594556/). *The Atlantic*. Retrieved July 25, 2019. "When he tried to match the air-to-ground transcript to an audiovisual recording, he found that the transcript was behind—and that one of the records had to be wrong about the time of Armstrong's first step."
137. Plimpton, George (December 1983). "Neil Armstrong's Famous First Words". *Esquire*. pp. 113–118.
138. "Apollo 11 Post Flight Press Conference, 16 September 1969" (https://history.nasa.gov/ap11ann/FirstLunarLanding/ch-7.html). NASA. Archived (https://web.archive.org/web/20151016015011/http://history.nasa.gov/ap11ann/FirstLunarLanding/ch-7.html) from the original on October 16, 2015. Retrieved July 24, 2015. "Yes, I did think about it. It was not extemporaneous, neither was it planned. It evolved during the conduct of the flight and I decided what the words would be while we were on the lunar surface just prior to leaving the LM."
139. Gray, Richard (December 30, 2012). "Neil Armstrong's family reveal origins of 'one small step' line" (https://www.telegraph.co.uk/news/science/space/9770712/Neil-Armstrongs-family-reveal-origins-of-one-small-step-line.html). *The Daily Telegraph*. Archived (https://web.archive.org/web/20150701041241/http://www.telegraph.co.uk/news/science/space/9770712/Neil-Armstrongs-family-reveal-origins-of-one-small-step-line.html) from the original on July 1, 2015. Retrieved July 24, 2015.
140. Chaikin, Andrew (January 4, 2013). "Neil Armstrong Didn't Lie About 'One Small Step' Moon Speech, Historian Says" (http://www.space.com/19136-neil-armstrong-moon-speech-truth.html). *Space.com*. Purch. Archived (https://web.archive.org/web/20150724160705/http://www.space.com/19136-neil-armstrong-moon-speech-truth.html) from the original on July 24, 2015. Retrieved July 24, 2015.
141. Nickell 2008, p. 175.
142. Goddard, Jacqui (October 2, 2006). "One small word is one giant sigh of relief for Armstrong" (https://www.thetimes.com/uk/science/article/one-small-word-is-one-giant-sigh-of-relief-for-armstrong-8ffrjd0s6wz). *The Times*. London. Retrieved December 31, 2012.
143. Ford, Peter Shann (September 17, 2006). "Electronic Evidence and Physiological Reasoning Identifying the Elusive Vowel "a" in Neil Armstrong's Statement on First Stepping onto the Lunar Surface" (http://www.collectspace.com/news/news-100306a.html). *collectSPACE*. Archived (https://web.archive.org/web/20070927015424/http://www.collectspace.com/news/news-100306a.html) from the original on September 27, 2007. Retrieved August 28, 2007.

144. "Software Finds Missing 'a' in Armstrong's Moon Quote" (https://web.archive.org/web/20061004151135/http://www.cnn.com/2006/US/09/30/moon.quote.ap/index.html). CNN. Associated Press. October 1, 2006. Archived from the original (http://www.cnn.com/2006/US/09/30/moon.quote.ap/index.html) on October 4, 2006.
145. Smith, Veronica (October 2, 2006). "Armstrong's Moon landing speech rewritten" (https://web.archive.org/web/20070831202457/http://www.cosmosmagazine.com/node/717). *Cosmos*. Agence France-Presse. Archived from the original (http://www.cosmosmagazine.com/node/717) on August 31, 2007. Retrieved August 29, 2007.
146. Carreau, Mark (September 29, 2006). "High-tech analysis may rewrite space history" (http://www.chron.com/disp/story.mpl/front/4225505.html). *Houston Chronicle*. Archived (https://web.archive.org/web/20061004192255/http://www.chron.com/disp/story.mpl/front/4225505.html) from the original on October 4, 2006. Retrieved September 30, 2006.
147. Carreau, Mark (September 30, 2006). "Hear what Neil Armstrong really said on the moon" (https://www.chron.com/news/nation-world/article/Hear-what-Neil-Armstrong-really-said-on-the-moon-1862496.php). *Houston Chronicle*. Retrieved July 25, 2019.
148. Language Logs:
    - Beaver, David. "One small step backwards" (http://itre.cis.upenn.edu/%7Emyl/languagelog/archives/003630.html). *Language Log*. University of Pennsylvania. Archived (https://web.archive.org/web/20171108072803/http://itre.cis.upenn.edu/%7Emyl/languagelog/archives/003630.html) from the original on November 8, 2017. Retrieved February 28, 2018. (including audio)
    - Liberman, Mark. "One 75-millisecond step before a "man" " (http://itre.cis.upenn.edu/%7Emyl/languagelog/archives/003632.html). *Language Log*. University of Pennsylvania. Archived (https://web.archive.org/web/20171108072810/http://itre.cis.upenn.edu/%7Emyl/languagelog/archives/003632.html) from the original on November 8, 2017. Retrieved February 28, 2018.
    - Beaver, David. "Armstrong's abbreviated article: the smoking gun?" (http://itre.cis.upenn.edu/~myl/languagelog/archives/003634.html). *Language Log*. University of Pennsylvania. Archived (https://web.archive.org/web/20171108072811/http://itre.cis.upenn.edu/~myl/languagelog/archives/003634.html) from the original on November 8, 2017. Retrieved February 28, 2018.
    - Beaver, David. "Armstrong's abbreviated article: notes from the expert" (http://itre.cis.upenn.edu/%7Emyl/languagelog/archives/003635.html). *Language Log*. University of Pennsylvania. Archived (https://web.archive.org/web/20171108072827/http://itre.cis.upenn.edu/%7Emyl/languagelog/archives/003635.html) from the original on November 8, 2017. Retrieved February 28, 2018.
    - Beaver, David. "First Korean on the moon!" (http://itre.cis.upenn.edu/~myl/languagelog/archives/003639.html). *Language Log*. University of Pennsylvania. Archived (https://web.archive.org/web/20171108072825/http://itre.cis.upenn.edu/~myl/languagelog/archives/003639.html) from the original on November 8, 2017. Retrieved February 28, 2018.
    - Liberman, Mark. "What Neil Armstrong said" (http://itre.cis.upenn.edu/~myl/languagelog/archives/003645.html). *Language Log*. University of Pennsylvania. Archived (https://web.archive.org/web/20171108072829/http://itre.cis.upenn.edu/~myl/languagelog/archives/003645.html) from the original on November 8, 2017. Retrieved February 28, 2018.
149. Baese-Berk, M. M.; Dilley, L. C.; Schmidt, S.; Morrill, T. H.; Pitt, M. A. (2016). "Revisiting Neil Armstrong's Moon-Landing Quote: Implications for Speech Perception, Function Word Reduction, and Acoustic Ambiguity" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5014323). *PLOS ONE*. **11** (9): 1–11. Bibcode:2016PLoSO..1155975B (https://ui.adsabs.harvard.edu/abs/2016PLoSO..1155975B). doi:10.1371/journal.pone.0155975 (https://doi.org/10.1371%2Fjournal.pone.0155975). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 5014323 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5014323). PMID 27603209 (https://pubmed.ncbi.nlm.nih.gov/27603209).

150. Jones, Eric M. "One Small Step, time 109:24:23" (https://www.hq.nasa.gov/office/pao/History/alsj/a11/a11.step.html). *Apollo 11 Surface Journal*. NASA. Archived (https://web.archive.org/web/20130102060848/http://www.hq.nasa.gov/office/pao/History/alsj/a11/a11.step.html) from the original on January 2, 2013. Retrieved December 18, 2012.
151. "Apollo 11 Mission Overview" (https://www.nasa.gov/mission_pages/apollo/missions/apollo11.html). NASA. April 17, 2015. Archived (https://web.archive.org/web/20180209204039/https://www.nasa.gov/mission_pages/apollo/missions/apollo11.html) from the original on February 9, 2018. Retrieved March 4, 2018.
152. Stuckey, Alex (October 29, 2018). " 'Giant leap' for space collectibles: Neil Armstrong's personal collection goes on auction block" (https://www.chron.com/news/nation-world/article/Giant-leap-for-space-collectibles-Neil-13345916.php). *Houston Chronicle*. Associated Press. Retrieved May 26, 2019.
153. "Total Population—Both Sexes" (https://web.archive.org/web/20170730155936/https://esa.un.org/unpd/wpp/DVD/Files/1_Indicators%20(Standard)/EXCEL_FILES/1_Population/WPP2017_POP_F01_1_TOTAL_POPULATION_BOTH_SEXES.xlsx). United Nations Population Division. Archived from the original (https://esa.un.org/unpd/wpp/DVD/Files/1_Indicators%20(Standard)/EXCEL_FILES/1_Population/WPP2017_POP_F01_1_TOTAL_POPULATION_BOTH_SEXES.xlsx) (xlsx) on July 30, 2017. Retrieved April 7, 2018.
154. Chaikin 2007, p. 212.
155. Johnson 2008, p. 60.
156. Hansen 2012, pp. 503–504.
157. Hansen 2005, pp. 505–506.
158. Jones, Eric M. (July 28, 2011). "AS11-40-5886" (https://web.archive.org/web/20110728042723/http://next.nasa.gov/alsj/a11/a11.5886.html). *Apollo 11 Lunar Surface Journal*. NASA. Archived from the original (https://next.nasa.gov/alsj/a11/a11.5886.html) on July 28, 2011. Retrieved May 13, 2011.
159. Jones, Eric M.; Glover, Ken (1995). "EASEP Deployment and Closeout" (https://www.hq.nasa.gov/alsj/a11/a11.clsout.html). *Apollo 11 Lunar Surface Journal*. NASA. 111:36:38. Archived (https://web.archive.org/web/20140225025455/http://www.hq.nasa.gov/alsj/a11/a11.clsout.html) from the original on February 25, 2014. Retrieved March 28, 2014.
160. "Summary Data on Apollo Missions" (https://web.archive.org/web/20120117134054/http://history.nasa.gov/SP-4214/app5.html). NASA. Archived from the original (https://history.nasa.gov/SP-4214/app5.html) on January 17, 2012. Retrieved May 20, 2011.
161. "Neil Armstrong Explains His Famous Apollo 11 Moonwalk" (http://www.space.com/10469-neil-armstrong-explains-famous-apollo-11-moonwalk.html). *Space.com*. December 10, 2010. Archived (https://web.archive.org/web/20130802234446/http://www.space.com/10469-neil-armstrong-explains-famous-apollo-11-moonwalk.html) from the original on August 2, 2013. Retrieved October 14, 2013.
162. Hansen 2005, pp. 489–490.
163. "Apollo 11 Command and Service Module (CSM)" (https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1969-059A). NASA. Retrieved August 26, 2012.
164. "Apollo 11 Crew Starts World Tour" (https://www.newspapers.com/clip/27494178/the_logan_daily_news/). *Logan Daily News*. Logan, Ohio. Associated Press. September 29, 1969. p. 1 – via Newspapers.com.
165. Nixon, Richard (August 13, 1969). Peters, Gerhard; Woolley, John T. (eds.). "Remarks at a Dinner in Los Angeles Honoring the Apollo 11 Astronauts" (https://www.presidency.ucsb.edu/documents/remarks-dinner-los-angeles-honoring-the-apollo-11-astronauts). *The American Presidency Project*. University of California, Santa Barbara. Retrieved July 20, 2019.
166. "President Offers Toast to 'Three Brave Men' " (https://www.newspapers.com/clip/27531303/the_evening_sun/). *The Evening Sun*. Baltimore, Maryland. Associated Press. August 14, 1969. p. 1 – via Newspapers.com.

167. Smith, Merriman (August 14, 1969). "Astronauts Awed by the Acclaim" (https://www.newspapers.com/clip/27531557/the_honolulu_advertiser/). *The Honolulu Advertiser*. United Press International. p. 1 – via Newspapers.com.
168. Hansen 2005, p. 580.
169. Hansen 2005, p. 583.
170. Hansen 2005, pp. 582–584.
171. Riley, Christopher (July 10, 2009). "The Moon Walkers: Twelve Men Who Have Visited Another World" (https://www.theguardian.com/science/2009/jul/09/apollo-astronauts-walking-moon). *The Guardian*. Archived (https://web.archive.org/web/20140204055219/http://www.theguardian.com/science/2009/jul/09/apollo-astronauts-walking-moon) from the original on February 4, 2014. Retrieved May 3, 2011.
172. Hansen 2005, p. 584.
173. "Neil Armstrong Remembered" (https://web.archive.org/web/20151208165113/http://ceas.uc.edu/about/neil-armstrong-remembered.html). *University of Cincinnati*. Archived from the original (http://ceas.uc.edu/about/neil-armstrong-remembered.html) on December 8, 2015. Retrieved November 28, 2015.
174. Hansen 2012, p. 590.
175. Jones, Eric M. (November 1, 2005). "Apollo 11 Crew Information" (https://web.archive.org/web/20070828070600/http://www.hq.nasa.gov/alsj/a11/a11.crew.html). *Apollo 11 Lunar Surface Journal*. NASA. Archived from the original (https://www.hq.nasa.gov/alsj/a11/a11.crew.html) on August 28, 2007. Retrieved August 28, 2007.
176. Hansen 2005, pp. 590–594.
177. Hansen 2005, pp. 600–603.
178. Hansen 2005, pp. 610–616.
179. Hansen 2012, pp. 609–610.
180. Hansen 2012, pp. 616–617.
181. Hansen 2005, p. 595.
182. Hansen 2005, p. 596.
183. Hansen 2005, pp. 596–598.
184. "EDO Corporation CEO James M. Smith to become Chairman upon retirement of Neil A. Armstrong" (https://web.archive.org/web/20061017094755/http://www.edocorp.com/pr2002/02r0208.htm) (Press release). EDO Corporation. February 8, 2000. Archived from the original (http://www.edocorp.com/pr2002/02r0208.htm) on October 17, 2006. Retrieved July 1, 2006.
185. Colavito, Jason. "Remembering Neil Armstrong's Brush with Ancient Astronauts" (https://www.jasoncolavito.com/blog/remembering-neil-armstrongs-brush-with-ancient-astronauts). *JASON COLAVITO*. Retrieved May 2, 2022.
186. Judson, David (November 1976). "Los Tayos Expedition, Ecuador : July-August 1976". *British Cave Research Association*. **14**: 26–30.
187. Hansen 2012, p. 609.
188. "Neil Armstrong, first man on the Moon, dead at 82" (https://www.thenationalnews.com/uae/neil-armstrong-first-man-on-the-moon-dead-at-82-1.607011). *The National*. August 26, 2012. Archived (https://web.archive.org/web/20180304172524/https://www.thenational.ae/uae/neil-armstrong-first-man-on-the-moon-dead-at-82-1.607011) from the original on March 4, 2018. Retrieved February 28, 2018.
189. Granath, Bob (August 30, 2012). "Neil Armstrong Praised as a Reluctant American Hero" (https://web.archive.org/web/20180902042741/https://www.nasa.gov/content/neil-armstrong-praised-as-a-reluctant-american-hero/). NASA. Archived from the original (https://www.nasa.gov/content/neil-armstrong-praised-as-a-reluctant-american-hero) on September 2, 2018. Retrieved July 8, 2018.

190. Hampton, Olivia (August 25, 2012). "Neil Armstrong a 'reluctant American hero': family" (https://www.theglobeandmail.com/news/world/neil-armstrong-a-reluctant-american-hero-family/article4499479/). *The Globe and Mail*. Retrieved July 8, 2018.
191. Seymour, Gene (August 27, 2012). "Neil Armstrong, a hero who shunned fame" (https://www.cnn.com/2012/08/26/opinion/seymour-armstrong-appreciation/index.html). CNN. Retrieved June 9, 2018.
192. Farhi, Paul (July 20, 2009). "Neil Armstrong Took One Small Step, Then Made a Giant Retreat Into Private Life" (https://www.washingtonpost.com/wp-dyn/content/article/2009/07/19/AR2009071901771.html). *The Washington Post*. Retrieved June 8, 2018.
193. "Tributes paid to Neil Armstrong, the humblest of American heroes" (https://www.telegraph.co.uk/news/science/space/9500380/Tributes-paid-to-Neil-Armstrong-the-humblest-of-American-heroes.html). *The Daily Telegraph*. August 26, 2013. Archived (https://ghostarchive.org/archive/20220110/https://www.telegraph.co.uk/news/science/space/9500380/Tributes-paid-to-Neil-Armstrong-the-humblest-of-American-heroes.html) from the original on January 10, 2022. Retrieved July 8, 2018.
194. Shapiro 2012, pp. 9, 267, 268.
195. Chaikin 2007, pp. 568–570.
196. Hart, Hugh (March 9, 2010). "NASA Footage Sets Scene for Quantum Quest Movie" (https://www.wired.com/2010/09/quantum-quest/). *Wired*. Archived (https://web.archive.org/web/20180301224818/https://www.wired.com/2010/09/quantum-quest/) from the original on March 1, 2018. Retrieved February 28, 2018.
197. "Quantum Quest" (https://web.archive.org/web/20130108191010/http://jupiter9productions.com/News.aspx). jupiter9productions.com. Archived from the original (http://jupiter9productions.com/news.aspx) on January 8, 2013. Retrieved October 8, 2012.
198. Anson, Robert Sam (November 2000). "Birth of an MTV Nation" (https://www.vanityfair.com/culture/features/2000/11/mtv200011?printable=true&currentPage=allPittman). *Vanity Fair*. Archived (https://web.archive.org/web/20141227164327/http://www.vanityfair.com/culture/features/2000/11/mtv200011?printable=true&currentPage=allPittman) from the original on December 27, 2014. Retrieved March 4, 2018.
199. Hansen 2005, p. 628.
200. "Neil Armstrong, Hallmark Settle" (https://www.chicagotribune.com/1995/12/02/neil-armstrong-hallmark-settle/). *Chicago Tribune*. December 2, 1995. Archived (https://web.archive.org/web/20110103194725/http://articles.chicagotribune.com/1995-12-02/business/9512020040_1_neil-armstrong-hallmark-cards-ornament) from the original on January 3, 2011. Retrieved May 19, 2011.
201. Hansen 2012, pp. 622–623.
202. Hansen 2005, pp. 600–601.
203. Hansen 2005, p. 33.
204. Hansen 2005, p. 35.
205. Congressional Record (Bound Edition). (September 16–22, 1969.) Volume 115. Part 19. p. 25611. US Government Printing Office. ("Joint Meeting of the Two Houses of Congress to Receive the Apollo 11 Astronauts". September 16, 1969.)
206. Hansen 2005, p. 575.
207. Abramson 2004, p. 93.
208. Hansen 2005, pp. 630–631.
209. Johnston, Willie (July 20, 2009). "Recalling Moon man's 'muckle' leap" (http://news.bbc.co.uk/2/hi/uk_news/scotland/8158762.stm). BBC News. Archived (https://web.archive.org/web/20090720100435/http://news.bbc.co.uk/2/hi/uk_news/scotland/8158762.stm) from the original on July 20, 2009. Retrieved July 20, 2009.
210. Hansen 2005, p. 13.

211. "Neil Armstrong and the Pickle" (https://web.archive.org/web/20190705060523/http://www.evergreensoaring.info/content/neil-armstrong-and-pickle). Evergreen Soaring. Archived from the original (http://www.evergreensoaring.info/content/neil-armstrong-and-pickle) on July 5, 2019. Retrieved July 6, 2019.
212. Hansen 2005, p. 608.
213. Hansen 2005, pp. 639–640.
214. Stein, Ellin (October 15, 2018). "What's Fact and What's Fiction in First Man, the New Neil Armstrong Movie" (https://slate.com/culture/2018/10/first-man-fact-vs-fiction-neil-armstrong-movie-daughter-bracelet.html). *Slate*. Retrieved October 21, 2018.
215. Schorn, Daniel (July 2, 2006). "Being The First Man On The Moon" (https://www.cbsnews.com/news/being-the-first-man-on-the-moon/). CBS News. Archived (https://web.archive.org/web/20101013013459/http://www.cbsnews.com/stories/2005/11/03/60minutes/main1008288_page3.shtml) from the original on October 13, 2010. Retrieved January 9, 2011.
216. Johnston, John; Amrhein, Saundra; Thompson, Richelle (July 18, 1999). "Neil Armstrong, Reluctant Hero" (https://web.archive.org/web/20200518093111/http://www.enquirer.com/editions/1999/07/18/loc_neil_armstrong_the.html/). *The Cincinnati Enquirer*. Archived from the original (http://www.enquirer.com/editions/1999/07/18/loc_neil_armstrong_the.html) on May 18, 2020. Retrieved March 4, 2018.
217. Hansen 2005, pp. 643–645.
218. "Armstrong, barber in fight over locks" (https://web.archive.org/web/20121106001601/http://www.highbeam.com/doc/1P2-11945915.html). *Dayton Daily News*. Dayton, OH. June 1, 2005. Archived from the original on November 6, 2012. Retrieved May 17, 2011 – via HighBeam Research.
219. Hansen 2012, p. 628.
220. "Original Moonwalker: How Neil Armstrong kept his feet on the ground despite becoming the most famous man on Earth" (https://www.independent.co.uk/news/science/original-moonwalker-how-neil-armstrong-kept-his-feet-on-the-ground-despite-becoming-the-most-famous-1727870.html). *The Independent*. July 2, 2009. Archived (https://ghostarchive.org/archive/20220618/https://www.independent.co.uk/news/science/original-moonwalker-how-neil-armstrong-kept-his-feet-on-the-ground-despite-becoming-the-most-famous-1727870.html) from the original on June 18, 2022. Retrieved July 8, 2018.
221. Shane, Scott; Kiliff, Sarah (July 23, 2019). "Neil Armstrong's Death, and a Stormy, Secret $6 Million Settlement" (https://www.nytimes.com/2019/07/23/us/neil-armstrong-wrongful-death-settlement.html). *The New York Times*. Retrieved July 23, 2019.
222. "Armstrong, First Man on the Moon, Recovering From Heart Surgery" (https://www.reuters.com/article/us-usa-people-armstrong-idUSBRE8771KU20120808). *Reuters*. August 8, 2012. Archived (https://web.archive.org/web/20120809070811/http://www.reuters.com/article/2012/08/08/us-usa-people-armstrong-idUSBRE8771KU20120808) from the original on August 9, 2012. Retrieved August 8, 2012.
223. H. Maugh II, Thomas (August 9, 2012). "Neil Armstrong recovering well after cardiac bypass surgery" (https://www.latimes.com/science/la-xpm-2012-aug-09-la-sci-sn-armstrong-heart-surgery-20120809-story.html). *Los Angeles Times*. Archived (https://web.archive.org/web/20120921021918/http://articles.latimes.com/2012/aug/09/science/la-sci-sn-armstrong-heart-surgery-20120809) from the original on September 21, 2012. Retrieved January 5, 2014.
224. Stone, Judy (September 3, 2012). "Neil Armstrong's Death – a Medical Perspective" (http://blogs.scientificamerican.com/molecules-to-medicine/2012/09/03/neil-armstrongs-deatha-medical-perspective/). *Scientific American*. Archived (https://web.archive.org/web/20120904034718/http://blogs.scientificamerican.com/molecules-to-medicine/2012/09/03/neil-armstrongs-deatha-medical-perspective/) from the original on September 4, 2012. Retrieved December 30, 2012.

225. "Space legend Neil Armstrong dies" (http://www.cnn.com/2012/08/25/us/neil-armstrong-obit/index.html). CNN. August 25, 2012. Archived (https://web.archive.org/web/20121229221442/http://www.cnn.com/2012/08/25/us/neil-armstrong-obit/index.html) from the original on December 29, 2012. Retrieved January 30, 2022.
226. "President Obama's Statement on Neil Armstrong's Death" (https://blogs.wsj.com/washwire/2012/08/25/president-obamas-statement-on-neil-armstrongs-death/). *The Wall Street Journal*. August 25, 2012. Archived (https://web.archive.org/web/20120829012207/http://blogs.wsj.com/washwire/2012/08/25/president-obamas-statement-on-neil-armstrongs-death/) from the original on August 29, 2012. Retrieved August 26, 2012.
227. Wilford, John Noble (August 25, 2012). "Neil Armstrong, First Man on Moon, Dies at 82" (https://www.nytimes.com/2012/08/26/science/space/neil-armstrong-dies-first-man-on-moon.html). *The New York Times*. Associated Press. Archived (https://web.archive.org/web/20120825202024/http://www.nytimes.com/2012/08/26/science/space/neil-armstrong-dies-first-man-on-moon.html) from the original on August 25, 2012. Retrieved August 25, 2012.
228. "Statement by the President on the Passing of Neil Armstrong" (https://obamawhitehouse.archives.gov/the-press-office/2012/08/25/statement-president-passing-neil-armstrong). The White House, Office of the Press Secretary. August 25, 2012. Archived (https://web.archive.org/web/20170121094421/https://obamawhitehouse.archives.gov/the-press-office/2012/08/25/statement-president-passing-neil-armstrong) from the original on January 21, 2017. Retrieved August 26, 2012.
229. "Family Statement Regarding the Death of Neil Armstrong" (https://www.nasa.gov/home/hqnews/2012/aug/HQ_12_600_armstrong_family.html). NASA. August 25, 2012. Archived (https://web.archive.org/web/20121006100820/http://www.nasa.gov/home/hqnews/2012/aug/HQ_12_600_armstrong_family.html) from the original on October 6, 2012. Retrieved August 26, 2012.
230. Aldrin, Buzz [@TheRealBuzz] (August 25, 2012). "Buzz Aldrin on passing of Neil Armstrong" (https://twitter.com/TheRealBuzz/status/239488144288927744) (Tweet). Retrieved August 25, 2012 – via Twitter.
231. Aldrin, Buzz (August 25, 2012). "Buzz Aldrin's Official Statement on the Passing of Neil Armstrong" (http://buzzaldrin.com/buzz-aldrins-official-statement-on-the-passing-of-neil-armstrong/). Archived (https://web.archive.org/web/20120830154317/http://buzzaldrin.com/buzz-aldrins-official-statement-on-the-passing-of-neil-armstrong/) from the original on August 30, 2012. Retrieved August 25, 2012.
232. "Neil Armstrong: 1930–2012" (https://web.archive.org/web/20121002103013/http://www.nasa.gov/topics/people/features/armstrong_obit.html). NASA. August 25, 2012. Archived from the original (https://www.nasa.gov/topics/people/features/armstrong_obit.html) on October 2, 2012. Retrieved August 25, 2012.
233. Collins, Michael (September 12, 2012). "The Neil Armstrong I knew – and flew with" (https://www.washingtonpost.com/opinions/michael-collins-the-neil-armstrong-i-knew--and-flew-with/2012/09/12/b3f7556c-fb7c-11e1-8adc-499661afe377_story.html). *The Washington Post*. Archived (https://web.archive.org/web/20120913103748/http://www.washingtonpost.com/opinions/michael-collins-the-neil-armstrong-i-knew--and-flew-with/2012/09/12/b3f7556c-fb7c-11e1-8adc-499661afe377_story.html) from the original on September 13, 2012. Retrieved September 13, 2012.
234. "NASA Administrator Statement on Neil Armstrong's Death" (https://www.nasa.gov/home/hqnews/2012/aug/HQ_12-601_Bolden_Statement.html). NASA. August 25, 2012. Archived (https://web.archive.org/web/20120826070237/http://www.nasa.gov/home/hqnews/2012/aug/HQ_12-601_Bolden_Statement.html) from the original on August 26, 2012. Retrieved August 25, 2012.
235. "US space pioneer Neil Armstrong dies at 82 – Americas" (http://www.aljazeera.com/news/americas/2012/08/201282519933640323.html). Al Jazeera. Archived (https://web.archive.org/web/20120827025712/http://www.aljazeera.com/news/americas/2012/08/201282519933640323.html) from the original on August 27, 2012. Retrieved August 26, 2012.

236. Terrett, John (September 15, 2012). "Above the stars now" (http://blogs.aljazeera.com/blog/americas/above-stars-now). Al Jazeera. Archived (https://web.archive.org/web/20150120202904/http://blogs.aljazeera.com/blog/americas/above-stars-now) from the original on January 20, 2015. Retrieved September 26, 2012.
237. Allen, Nick (August 26, 2012). "Neil Armstrong: Barack Obama under pressure to grant state funeral" (https://web.archive.org/web/20120826202941/http://www.telegraph.co.uk/science/space/9501034/Neil-Armstrong-Barack-Obama-under-pressure-to-grant-state-funeral.html). *The Telegraph*. London. Archived from the original (https://www.telegraph.co.uk/science/space/9501034/Neil-Armstrong-Barack-Obama-under-pressure-to-grant-state-funeral.html) on August 26, 2012.
238. "Neil Armstrong Laid to Rest in Atlantic" (https://web.archive.org/web/20120913211523/http://www.nasa.gov/topics/people/features/armstrong_cathedral_memorial.html). NASA. Archived from the original (https://www.nasa.gov/topics/people/features/armstrong_cathedral_memorial.html) on September 13, 2012. Retrieved September 14, 2012.
239. Jackson, David (August 27, 2012). "Obama orders flags at half-staff for Neil Armstrong" (http://content.usatoday.com/communities/theoval/post/2012/08/obama-orders-flags-at-half-staff-for-armstrong/1). *USA Today*. Archived (https://web.archive.org/web/20120830003341/https://content.usatoday.com/communities/theoval/post/2012/08/obama-orders-flags-at-half-staff-for-armstrong/1) from the original on August 30, 2012. Retrieved January 13, 2013.
240. Shane, Scott; Kliff, Sarah (August 1, 2019). "A Scoop About Neil Armstrong Arrived in a Plain Brown Envelope" (https://www.nytimes.com/2019/08/01/reader-center/neil-armstrong-settlement-anonymous-tip.html). *The New York Times*. Retrieved September 21, 2020.
241. Shane, Scott; Kliff, Sarah; Craig, Susanne (July 27, 2019). " 'Would Dad Approve?' Neil Armstrong's Heirs Divide Over a Lucrative Legacy" (https://www.nytimes.com/2019/07/27/us/neil-armstrong-heirs.html). *The New York Times*. Retrieved July 27, 2019.
242. Apollo 12 First Steps (https://www.youtube.com/watch?v=YEEIJYrXn9s) on YouTube
243. Smith, Merriman (August 14, 1969). "Astronauts Awed by the Acclaim" (https://www.newspapers.com/clip/27531557/the_honolulu_advertiser/). *The Honolulu Advertiser*. United Press International. p. 1 – via Newspapers.com.
244. "The Cullum Geographical Medal" (https://web.archive.org/web/20170630023922/http://americangeo.org/the-cullum-geographical-medal/). *American Geographical Society*. Archived from the original (http://americangeo.org/the-cullum-geographical-medal/) on June 30, 2017.
245. "Collier Trophy – Collier 1960–1969 Winners" (https://web.archive.org/web/20070927130814/http://www.naa.aero/html/awards/index.cfm?cmsid=153). National Aeronautics Association. Archived from the original (http://www.naa.aero/html/awards/index.cfm?cmsid=153) on September 27, 2007.
246. "Agnew Confers Awards on Crews of 3 Apollos" (https://www.newspapers.com/clip/44062092/arizona_republic/). *Arizona Republic*. Phoenix, Arizona. Associated Press. November 14, 1970. p. 23 – via Newspapers.com.
247. "Past Goddard Trophy Winners" (https://web.archive.org/web/20120304104512/http://www.spaceclub.org/events/trophy.html). National Space Club. Archived from the original (http://www.spaceclub.org/events/trophy.html) on March 4, 2012. Retrieved January 27, 2013.
248. "Sylvanus Thayer Award Recipients" (https://www.westpointaog.org/ThayerAwardrecipients). West Point Association of Graduates. Archived (https://web.archive.org/web/20171012101647/https://www.westpointaog.org/ThayerAwardrecipients) from the original on October 12, 2017. Retrieved February 28, 2018.
249. "Wright Bros. 2000–2009 Recipients" (https://naa.aero/awards/awards-and-trophies/wright-brothers-memorial-trophy/wright-bros-2000-2009-winners). National Aeronautic Association. Archived (https://web.archive.org/web/20171208122457/https://naa.aero/awards/awards-and-trophies/wright-brothers-memorial-trophy/wright-bros-2000-2009-winners) from the original on December 8, 2017. Retrieved December 7, 2017.

250. "NASA Legends Awarded Congressional Gold Medal" (https://web.archive.org/web/20170519175531/https://www.nasa.gov/topics/people/features/gold_medal.html). NASA. November 16, 2011. Archived from the original (https://www.nasa.gov/topics/people/features/gold_medal.html) on May 19, 2017. Retrieved February 28, 2018.
251. "Mr. Neil Alden Armstrong" (https://nae.edu/29763/Mr-Neil-Alden-Armstrong). *NAE Website*.
252. "APS Member History" (https://search.amphilsoc.org/memhist/search?creator=Neil+Armstrong&title=&subject=&subdiv=&mem=&year=&year-max=&dead=&keyword=&smode=advanced). *search.amphilsoc.org*. Retrieved October 14, 2021.
253. "Apollo 11 astronauts honored for 'astonishing' mission" (https://web.archive.org/web/20180301164519/http://edition.cnn.com/TECH/space/9907/20/apollo.11.awards/). CNN. July 20, 1999. Archived from the original (http://edition.cnn.com/TECH/space/9907/20/apollo.11.awards/) on March 1, 2018. Retrieved February 28, 2018.
254. "NASA Honors Neil Armstrong With Exploration Award" (https://web.archive.org/web/20150602190438/http://www.nasa.gov/vision/space/features/armstrong_ambassador_of_exploration.html). NASA. April 18, 2006. Archived from the original (https://www.nasa.gov/vision/space/features/armstrong_ambassador_of_exploration.html) on June 2, 2015. Retrieved December 5, 2014.
255. "Neil Armstrong and Sally Ride Are 2013 General James E. Hill Lifetime Space Achievement Award Honorees" (https://web.archive.org/web/20130603040641/http://www.spacefoundation.org/media/press-releases/neil-armstrong-and-sally-ride-are-2013-general-james-e-hill-lifetime-space). Space Foundation. Archived from the original (http://www.spacefoundation.org/media/press-releases/neil-armstrong-and-sally-ride-are-2013-general-james-e-hill-lifetime-space) on June 3, 2013. Retrieved March 26, 2013.
256. "Honorees" (https://web.archive.org/web/20110519062713/http://cityoflancasterca.org/index.aspx?page=193). City of Lancaster. Archived from the original (http://www.cityoflancasterca.org/index.aspx?page=193) on May 19, 2011. Retrieved May 19, 2011.
257. Chandler, John (September 17, 1991). "Neil Armstrong to Join Lancaster Walk of Honor" (https://www.newspapers.com/clip/36299181/the_los_angeles_times/). *Los Angeles Times*. p. B3 – via Newspapers.com.
258. Locke, Robert (October 6, 1976). "Space Pioneers Enshrined" (https://www.newspapers.com/clip/29967024/las_vegas_optic/). *Las Vegas Optic*. Las Vegas, New Mexico. Associated Press. p. 6 – via Newspapers.com.
259. Clark, Amy (March 14, 1993). "Activities Honor Gemini Astronauts" (https://www.newspapers.com/clip/33582881/florida_today/). *Florida Today*. Cocoa, Florida. p. 41 – via Newspapers.com.
260. "Neil A. Armstrong" (https://web.archive.org/web/20111001025716/http://www.astronautscholarship.org/armstrong.html). Astronaut Scholarship Foundation. Archived from the original (http://www.astronautscholarship.org/armstrong.html) on October 1, 2011. Retrieved May 19, 2011.
261. Kirk, Amy (March 10, 2010). "Astronaut Legend Receives Naval Astronaut Wings Aboard 'Ike' " (https://web.archive.org/web/20180228041525/http://www.navy.mil/submit/display.asp?story_id=51836). United States Navy. Archived from the original (http://www.navy.mil/submit/display.asp?story_id=51836) on February 28, 2018. Retrieved February 28, 2018.
262. "Discovery Circumstances: Numbered Minor Planets (5001)-(10000)" (http://www.minorplanetcenter.net/iau/lists/NumberedMPs005001.html). The International Astronomical Minor Planet Center. Retrieved November 27, 2015.

263. "Search for Public School" (https://nces.ed.gov/ccd/schoolsearch/school_list.asp?Search=1&InstName=neil+armstrong&SchoolID=&Address=&City=&State=&Zip=&Miles=&County=&PhoneAreaCode=&Phone=&DistrictName=&DistrictID=&SchoolType=1&SchoolType=2&SchoolType=3&SchoolType=4&SpecificSchlTypes=all&IncGrade=-1&LoGrade=-1&HiGrade=-1). National Center for Educational Statistics. Archived (https://web.archive.org/web/20071011184021/http://nces.ed.gov/ccd/schoolsearch/school_list.asp?Search=1&InstName=neil+armstrong&SchoolID=&Address=&City=&State=&Zip=&Miles=&County=&PhoneAreaCode=&Phone=&DistrictName=&DistrictID=&SchoolType=1&SchoolType=2&SchoolType=3&SchoolType=4&SpecificSchlTypes=all&IncGrade=-1&LoGrade=-1&HiGrade=-1) from the original on October 11, 2007. Retrieved July 10, 2007.
264. "Ireland: What's in a name? Cold, hard cash" (https://web.archive.org/web/20110609215101/http://property.timesonline.co.uk/tol/life_and_style/property/article804378.ece). *The Times*. December 22, 2002. Archived from the original (http://property.timesonline.co.uk/tol/life_and_style/property/article804378.ece) on June 9, 2011. Retrieved August 28, 2007.
265. Knight, Andy (Winter 2000). "To the moon: Armstrong space museum offers history lessons on space travel" (https://web.archive.org/web/20071223011402/http://www.cincinnati.com/visitorsguide/stories/012800_moon.html). *Cincinnati.com*. Archived from the original (http://www.cincinnati.com/visitorsguide/stories/012800_moon.html) on December 23, 2007. Retrieved August 28, 2007.
266. "Auglaize County Neil Armstrong Airport" (http://www.neilarmstrongairport.com/). Auglaize County Neil Armstrong Airport. Archived (https://web.archive.org/web/20170921040243/http://neilarmstrongairport.com/) from the original on September 21, 2017. Retrieved March 4, 2018.
267. Vladykin, N. V.; Kovalenko, V. I.; Kashaev, A. A.; Sapozhnikov, A. N.; Pisarskaya, V. A. (1973). "A new silicate of calcium and zirconium – armstrongite". *Doklady Akademii Nauk SSSR*. **209**: 1185–1188.
268. Anderson, A. T. (1970). "Armalcolite, a new mineral from the Apollo 11 samples". *Geochimica et Cosmochimica Acta*. **34** (Supplement 1): 55–63. Bibcode:1970GeCAS...1...55A (https://ui.adsabs.harvard.edu/abs/1970GeCAS...1...55A).
269. Holsapple, Matt (October 16, 2004). "Purdue launching Neil Armstrong Hall for engineering's future" (http://news.uns.purdue.edu/html3month/2004/041016.Jischke.Armstrong.html). *Purdue University News*. Archived (https://web.archive.org/web/20071016105510/http://news.uns.purdue.edu/html3month/2004/041016.Jischke.Armstrong.html) from the original on October 16, 2007. Retrieved August 28, 2007.
270. Venere, Emil (October 27, 2007). "Neil Armstrong Hall is new home to Purdue engineering" (http://news.uns.purdue.edu/x/2007b/071027CelArmstrongDedication.html). *Purdue University News*. Archived (https://web.archive.org/web/20071230143412/http://news.uns.purdue.edu/x/2007b/071027CelArmstrongDedication.html) from the original on December 30, 2007. Retrieved January 5, 2008.
271. "An Act to Redesignate the Dryden Flight Research Center as the Neil A. Armstrong Flight Research Center and the Western Aeronautical Test Range as the Hugh L. Dryden Aeronautical Test Range" (https://purl.fdlp.gov/GPO/gpo46206). US Government Publishing Office. Retrieved February 28, 2018.
272. "Navy Announces Research Vessel to be Named in Honor of Neil Armstrong" (https://web.archive.org/web/20121202044715/http://www.navy.mil/submit/display.asp?story_id=69758). United States Navy. September 24, 2012. Archived from the original (http://www.navy.mil/submit/display.asp?story_id=69758) on December 2, 2012.
273. "Neil Armstrong Distinguished Visiting Fellows" (https://engineering.purdue.edu/NADVF). Purdue University. Retrieved April 30, 2019.
274. McGauley, John (October 14, 2005). "Discovering the Man Behind 'First Man' " (http://www.collectspace.com/news/news-013003b.html#100305). *collectSPACE*. Archived (https://web.archive.org/web/20180131140927/http://www.collectspace.com/news/news-013003b.html#100305) from the original on January 31, 2018. Retrieved February 28, 2018.

275. Yoong, Sean (September 6, 2005). "Neil Armstrong: Manned Mars mission 20 years away" (https://www.usatoday.com/tech/science/space/2005-09-06-mars-armstrong_x.htm). *USA Today*. Washington DC. Archived (https://web.archive.org/web/20110726194523/http://www.usatoday.com/tech/science/space/2005-09-06-mars-armstrong_x.htm) from the original on July 26, 2011. Retrieved May 3, 2011.
276. Stolworthy, Jacob (March 8, 2017). "First Man: Damien Chazelle and Ryan Gosling's Neil Armstrong biopic gets awards season release date" (https://www.independent.co.uk/arts-entertainment/films/news/ryan-gosling-damien-chazelle-neil-armstrong-biopic-first-man-release-date-trailer-oscars-2019-best-a7618146.html). *The Independent*. Archived (https://web.archive.org/web/20180201013746/http://www.independent.co.uk/arts-entertainment/films/news/ryan-gosling-damien-chazelle-neil-armstrong-biopic-first-man-release-date-trailer-oscars-2019-best-a7618146.html) from the original on February 1, 2018. Retrieved February 28, 2018.
277. Cornwell, Lisa (July 20, 2018). "One giant sale: Neil Armstrong's collection goes to auction" (https://phys.org/news/2018-07-giant-sale-neil-armstrong-auction.html). *phys.org*. Retrieved December 16, 2018.
278. "CAG-certified Armstrong Family Collection Tops $5.2 Million in First Auction" (https://www.pmgnotes.com/news/article/6960/). PMG. November 5, 2018. Retrieved December 16, 2018.
279. "Neil Armstrong's gold medal that flew to the Moon sold for more than $2 million" (https://www.foxnews.com/science/apollo-11-neil-armstrong-medal-moon-sold). *Fox News*. July 17, 2019. Retrieved July 31, 2025.
280. "Wright Flyer fabric lands at Purdue University Archives" (https://www.purdue.edu/newsroom/releases/2019/Q2/wright-flyer-fabric-lands-at-purdue-university-archives.html). *Purdue University News*. May 30, 2019. Retrieved July 5, 2019.
281. "Neil A. Armstrong papers, 1671-2012" (https://archives.lib.purdue.edu/repositories/2/resources/1158). *Purdue University Archives and Special Collections*. Archived (https://web.archive.org/web/20260218195128/https://archives.lib.purdue.edu/repositories/2/resources/1158) from the original on February 18, 2026. Retrieved May 29, 2026.
282. "Space Foundation Survey Reveals Broad Range of Space Heroes" (https://web.archive.org/web/20120723031011/http://www.spacefoundation.org/media/press-releases/space-foundation-survey-reveals-broad-range-space-heroes-early-astronauts-still?id=1038) (Press release). Space Foundation. October 27, 2010. Archived from the original (http://www.spacefoundation.org/news/story.php?id=1038) on July 23, 2012. Retrieved May 13, 2011.
283. "51 Heroes of Aviation" (http://www.flyingmag.com/photo-gallery/photos/51-heroes-aviation?pnid=41853). *Flying*. July 24, 2013. Archived (https://web.archive.org/web/20151016015011/http://www.flyingmag.com/photo-gallery/photos/51-heroes-aviation?pnid=41853) from the original on October 16, 2015. Retrieved February 28, 2018.
284. Kaplan, Jeremy A. (April 14, 2010). "Star Wars: Neil Armstrong, Obama Spar Over NASA's Future" (https://www.foxnews.com/science/star-wars-neil-armstrong-obama-spar-over-nasas-future). Fox News. Archived (https://web.archive.org/web/20151029081228/http://www.foxnews.com/scitech/2010/04/14/obama-vs-neil-armstrong-nasa-constellation/?test=latestnews) from the original on October 29, 2015. Retrieved February 28, 2018. "I suspect that even though the various questions are difficult and many, they are not as difficult and many as those we faced when we started the Apollo [space program] in 1961."
285. Armstrong, Neil; Lovell, James; Cernan, Eugene (April 14, 2010). "Armstrong: Obama NASA plan 'devastating' " (https://www.nbcnews.com/id/wbna36470363). NBC Nightly News. Retrieved October 18, 2018.
286. "Neil Armstrong wil nog best naar Mars" (http://www.nu.nl/wetenschap/2382331/neil-armstrong-wil-nog-best-mars.html) [Neil Armstrong still wants to go to Mars]. *NU.nl* (in Dutch). November 18, 2010. Archived (https://web.archive.org/web/20180220151853/https://www.nu.nl/wetenschap/2382331/neil-armstrong-wil-nog-best-mars.html) from the original on February 20, 2018. Retrieved February 28, 2018 – via ANP.

287. School District, Altoona Area. "Welcome to the Neil Armstrong Planetarium" (https://web.archive.org/web/20231103080035/https://aahs.aasdcat.com/apps/pages/index.jsp?uREC_ID=424697&type=d&pREC_ID=1034959). *Altoona Area High School*. Altoona Area School District. Archived from the original (https://aahs.aasdcat.com/apps/pages/index.jsp?uREC_ID=424697&type=d&pREC_ID=1034959) on November 3, 2023. Retrieved April 25, 2020.
288. Swanson, Conrad; Roeder, Tom (May 30, 2019). "Air Force Academy graduation notes: Class of 2019 gives nod to Neil Armstrong" (https://gazette.com/military/air-force-academy-graduation-notes-class-of-2019-gives-nod-to-neil-armstrong/article_c2647ece-8327-11e9-9c8a-13a69088cc68.html). *Colorado Springs Gazette*. Retrieved June 10, 2020.

References

- Abramson, Glenda (2004). *Religious Perspectives in Modern Muslim and Jewish Literatures*. Studies in Arabic and Middle-Eastern Literatures. London: Routledge. ISBN 978-0-415-35021-1. OCLC 56641703 (https://search.worldcat.org/oclc/56641703).
- Brooks, Courtney G.; Grimwood, James M.; Swenson, Loyd S. Jr. (2009) [1979]. *Chariots for Apollo: A History of Manned Lunar Spacecraft* (https://web.archive.org/web/20151020095653/http://www.hq.nasa.gov/office/pao/History/SP-4205/cover.html). The NASA History Series. Foreword by Samuel C. Phillips. Washington, DC: Scientific and Technical Information Branch, NASA. ISBN 978-0-486-46756-6. LCCN 79001042 (https://lccn.loc.gov/79001042). OCLC 227923959 (https://search.worldcat.org/oclc/227923959). NASA SP-4205. Archived from the original (https://www.hq.nasa.gov/office/pao/History/SP-4205/cover.html) on October 20, 2015. Retrieved August 1, 2013.
- Burgess, Colin (2013). *Moon Bound: Choosing and Preparing NASA's Lunar Astronauts*. Springer-Praxis books in space exploration. New York; London: Springer. ISBN 978-1-4614-3854-0. OCLC 905162781 (https://search.worldcat.org/oclc/905162781).
- Cernan, Eugene; Davis, Don (1999). *The Last Man on the Moon* (https://archive.org/details/lastmanonmoonast00cern_0). New York: St Martin's Griffin. ISBN 978-0-312-19906-7. OCLC 40200305 (https://search.worldcat.org/oclc/40200305).
- Chaikin, Andrew (1994). *A Man on the Moon: The Voyages of the Apollo Astronauts*. London: Penguin Books. ISBN 978-0-14-024146-4. OCLC 895935578 (https://search.worldcat.org/oclc/895935578).
- Chaikin, Andrew (2007) [1994]. *A Man on the Moon: The Voyages of the Apollo Astronauts*. London: Penguin Books. ISBN 978-0-14-311235-8. OCLC 958200469 (https://search.worldcat.org/oclc/958200469).
- Collins, Michael (2001). *Carrying the Fire: An Astronaut's Journeys*. New York: Cooper Square Press. ISBN 978-0-8154-1028-7. OCLC 45755963 (https://search.worldcat.org/oclc/45755963).
- Cortright, Edgar M. (1975). "Apollo Expeditions to the Moon". *NASA Special Publication*. **350**. Bibcode:1975NASSP.350.....C (https://ui.adsabs.harvard.edu/abs/1975NASSP.350.....C). OCLC 251694818 (https://search.worldcat.org/oclc/251694818). NASA SP 350.
- Cunningham, Walter (2010) [1977]. *The All-American Boys*. New York: ipicturebooks. ISBN 978-1-876963-24-8. OCLC 713908039 (https://search.worldcat.org/oclc/713908039).
- Hacker, Barton C.; Grimwood, James M. (2010) [1977]. *On the Shoulders of Titans: A History of Project Gemini* (https://ghostarchive.org/archive/20221009/http://history.nasa.gov/SP-4203.pdf) (PDF). NASA History Series. Washington, DC: NASA History Division, Office of Policy and Plans. ISBN 978-0-16-067157-9. OCLC 945144787 (https://search.worldcat.org/oclc/945144787). NASA SP-4203. Archived from the original (https://history.nasa.gov/SP-4203.pdf) (PDF) on October 9, 2022. Retrieved April 8, 2018.

- Hansen, James R. (2005). *First Man: The Life of Neil A. Armstrong*. New York: Simon & Schuster. ISBN 978-0-7432-5631-5. OCLC 937302502 (https://search.worldcat.org/oclc/937302502).
- Hansen, James R. (2012). *First Man: The Life of Neil A. Armstrong*. New York: Simon & Schuster. ISBN 978-1-4767-2781-3. OCLC 1029741947 (https://search.worldcat.org/oclc/1029741947).
- Harland, David (1999). *Exploring the Moon: The Apollo Expeditions*. London; New York: Springer. ISBN 978-1-85233-099-6. OCLC 982158259 (https://search.worldcat.org/oclc/982158259).
- Jenkins, Dennis R. (June 2000). *Hypersonics Before the Shuttle: A Concise History of the X-15 Research Airplane* (https://history.nasa.gov/monograph18.pdf) (PDF). Monographs in Aerospace History. Washington, DC: NASA. OCLC 421657636 (https://search.worldcat.org/oclc/421657636). NASA SP-2000-4518. Retrieved February 28, 2018.
- Johnson, Sandra L. (Fall 2008). "Red, White & Blue: U.S. Flag at Home on the Moon" (https://houstonhistorymagazine.org/wp-content/uploads/2014/03/red-white-and-blue-US-flag.pdf) (PDF). *Houston History Magazine*. **6** (1): 60. Retrieved March 17, 2014.
- Koestler-Grack, Rachel A. (2010). *Neil Armstrong* (https://archive.org/details/neilarmstrong0000koes). Pleasantville, New York: Gareth Stevens. ISBN 978-1-4339-2147-6. OCLC 300982973 (https://search.worldcat.org/oclc/300982973).
- Kraft, Chris (2001). *Flight: My Life in Mission Control* (https://archive.org/details/flight00chri). New York: Dutton. ISBN 978-0-525-94571-0. OCLC 49323520 (https://search.worldcat.org/oclc/49323520).
- Kranz, Gene (2000). *Failure is not an Option: Mission Control From Mercury to Apollo 13 and Beyond* (https://archive.org/details/isbn_9780743200790). Alexandria, Virginia: Simon & Schuster. ISBN 978-1-4391-4881-5. OCLC 810544750 (https://search.worldcat.org/oclc/810544750).
- Lovell, Jim; Kluger, Jeffrey (2000). *Apollo 13*. Boston: Houghton Mifflin. ISBN 978-0-618-05665-1. OCLC 1011809615 (https://search.worldcat.org/oclc/1011809615).
- Manned Spacecraft Center (November 1969). *Apollo 11 Mission Report* (https://web.archive.org/web/20030616190156/http://history.nasa.gov/alsj/a11/A11_MissionReport.pdf) (PDF). Houston: NASA. OCLC 8444918 (https://search.worldcat.org/oclc/8444918). MSC 00171. Archived from the original (https://history.nasa.gov/alsj/a11/A11_MissionReport.pdf) (PDF) on June 16, 2003. Retrieved February 28, 2018.
- Nelson, Craig (2009). *Rocket Men: The Epic Story of the First Men on the Moon* (https://archive.org/details/isbn_9780670021031). New York City: Viking. ISBN 978-0-7195-6948-7. OCLC 699213335 (https://search.worldcat.org/oclc/699213335).
- Nickell, Duane S. (2008). *Guidebook for the Scientific Traveler: Visiting Astronomy and Space*. New Brunswick, New Jersey: Rutgers University Press. ISBN 978-0-8135-4374-1. OCLC 190785292 (https://search.worldcat.org/oclc/190785292).
- Orloff, Richard W. (2000). *Apollo by the Numbers: A Statistical Reference* (https://history.nasa.gov/SP-4029.pdf) (PDF). NASA History Series. Washington, DC: NASA History Division, Office of Policy and Plans. ISBN 978-0-16-050631-4. LCCN 00061677 (https://lccn.loc.gov/00061677). OCLC 829406439 (https://search.worldcat.org/oclc/829406439). NASA SP-2000-4029. Retrieved March 4, 2018.
- Reichl, Eugen (2016). *Project Gemini*. Atglen, Pennsylvania: Schiffer. ISBN 978-0-7643-5070-2.
- Shapiro, Michael E. (2012). *Cable Television Prime Time Programming 1990–2010*. Jefferson, North Carolina: McFarland & Co. ISBN 978-0-7864-7087-7. OCLC 878810736 (https://search.worldcat.org/oclc/878810736).
- Smith, Andrew (2005). *Moondust: In Search of the Men Who Fell to Earth*. London: Bloomsbury. ISBN 978-0-7475-6368-6. OCLC 728066122 (https://search.worldcat.org/oclc/728066122).

Further reading

- Barbree, Jay (2014). *Neil Armstrong: A Life of Flight*. New York: Thomas Dunne Books. ISBN 978-1-250-04071-8. OCLC 900815422 (https://search.worldcat.org/oclc/900815422).
- French, Francis; Burgess, Colin (2010). *In the Shadow of the Moon*. Lincoln, Nebraska, and London: University of Nebraska Press. ISBN 978-0-8032-2979-2. OCLC 1019883802 (https://search.worldcat.org/oclc/1019883802).
- Thompson, Milton O. (1992). *At the Edge of Space: The X-15 Flight Program* (https://archive.org/details/atedgeofspacex1500thom). Washington, D.C.: Smithsonian Books. ISBN 978-1-56098-107-7. OCLC 925195868 (https://search.worldcat.org/oclc/925195868).

External videos

After Words interview with Jay Barbree on *Neil Armstrong: A Life of Flight*, July 19, 2014 (https://www.c-span.org/video/?320051-1/after-words-jay-barbree), C-SPAN

External links

- Neil Armstrong Commemorative Website (https://sites.libraries.uc.edu/armstrong/home)
- Neil Armstrong collected news and commentary (https://www.nytimes.com/topic/person/neil-armstrong). *The New York Times*.
- Neil Armstrong (https://www.theguardian.com/science/neil-armstrong) collected news and commentary at *The Guardian*
- Appearances (https://www.c-span.org/person/?8521) on C-SPAN

Retrieved from "https://en.wikipedia.org/w/index.php?title=Neil_Armstrong&oldid=1381325153"